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FOR THE SAFETY OF AIR NAVIGATION



EUROCONTROL EXPERIMENTAL CENTRE

USER MANUAL FOR THE BASE OF AIRCRAFT DATA (BADA) REVISION 3.11

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Abstract:

The Base of Aircraft Data (BADA) provides a set of ASCII files containing performance and operating procedure coefficients for 405 different aircraft types. The coefficients include those used to calculate thrust, drag and fuel flow and those used to specify nominal cruise, climb and descent speeds. User Manual for Revision 3.11 of BADA provides definitions of each of the coefficients and then explains the file formats. Instructions for remotely accessing the files via Internet are also given.

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SUMMARY

The Base of Aircraft Data (BADA) provides a set of ASCII files containing performance and operating procedure coefficients for 405 different aircraft types. The coefficients include those used to calculate thrust, drag and fuel flow and those used to specify nominal cruise, climb and descent speeds. The User Manual for Revision 3.11 of BADA provides definitions of each of the coefficients and then explains the file formats. Instructions for remotely accessing the files via Internet are also given.

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USER MANUAL MODIFICATION HISTORY

Issue Number	Release Date	Comments
Revision 2.1 Issue 1.0	31.05.94	First release of document
Revision 2.2 Issue 1.0	25.01.95	Released with BADA Revision 2.2 - 8 new aircraft models - 2 modified aircraft models - 2 modified equivalences - 6 removed equivalences - 14 new equivalences - modified file formats - additional Synonym File - corrections to formulas in previous version of document - additional description of total-energy and standard atmosphere equations
Revision 2.3 Issue 1.0	08.06.95	Released with BADA Revision 2.3 - document format modified to be consistent with EEC Technical Note standards - new A/C models for B73V and D328 - MD11 changed from equivalence to direct support - generic military fighter model, FGTR, replaces specific fighter models - maximum payload parameter added to all OPF files - Performance Tables Files (*.PTF) introduced - ISA equations used for TAS/CAS conversions instead of approximations (Section 3.2) - use only one formula for correction of speeds at mass values different from reference mass (Section 3.3) - add specification of minimum speed as function of stall speed (Section 3.4) - specification of transition altitude calculated added (Section 4.1) - speed schedules modified for climb (Section 4.1) and descent (Section 4.3) - modify Internet address for remote access and EUROCONTROL contact person (Section 6) - removed Section 7 (General Comments)

Issue Number	Release Date	Comments
Revision 2.4 Issue 1.0	04.01.96	Released with BADA Revision 2.4 - new A/C model for FK70 - C421 changed from equivalence to directly supported - 10 new equivalences - 1 modified equivalence - 3 re-developed models - introduction of dynamic maximum altitude - new temperature correction on thrust - modified max.alt for 4 models - modified minimum weight for 2 models - modified temperature coefficients for 12 models - esf calculation for constant CAS below tropopause changed from binomial approximation to exact formula - cruise Mach numbers changed for 4 models - change in altitude limit for descent speed
Revision 2.5 Issue 1.0	20.01.97	- re-developed models: EA32, B737, B73S, AT42, B767, DC9, BA46, FK10, MD80. - new model: CL65, DH83 - change of minimum speeds - change of climb/descent speed schedules - cruise fuel flow correction - buffeting speed for jet a/c - addition of BADA.GPF file - definition of acceleration limits, bank angles and holding speeds - 38 new equivalences added (SA4, SA5, SweDen 96) - 1 modified equivalence (B74S) - modified climb/cruise speeds (BE90, BE99, E120, PA42, FK50, B73F, B767,B747, B727, DA20) - Format changes in OPF file - Header changes in PTF file - Temperature influence on thrust limitation changed - Unit of Vstall in OPF file changed to KCAS - Correction of typing errors - Correction of APF file format explanation
Revision 2.6 Issue 1.0	01.09.97	- Added non-clean drag and thrust data for: EA32, B73S, MD80, B737, B747, FK10, AT42, B767 and CL65 models - All models mentioned above were re-developed using new clean drag data. - ND16, E120 and FK50 were re-modelled to correct the cruise speed capability. - Change of speed schedule in the take-off / initial climb

Issue Number	Release Date	Comments
		phase and approach / landing phase - Change in descent thrust algorithm - Use of exact formula for density below tropopause instead of approximation. - Addition of formula for pressure above tropopause - Change of buffeting limit to 1.2g (was 1.3g) - Change of OPF file format - Buffeting coefficients for B757 and MD80 were corrected. - Hmo for B747 model was corrected to 45,000 ft - Low altitude descent behaviour corrected for: SW3, PAYE, DA50, DA10, D328, C421, BE99, BE20 and BE90 models - Correction of some minor typing errors - dynamic maximum altitude coefficients changed for B747, B74F, C130 and EA30 - Saab 2000 (SB20) added as equivalent of D328 - Modified algorithm for lift coefficient
Revision 3.0 Issue 1.0	01.03.98	- Climb speed law changed for jet aircraft - Descent speed law changed for jet, turbo and piston - Reduced power climbs - B777, SB20 and B73X models were added - DA01 model was removed - Use of ICAO doc. 8643/25 standard, which resulted in the removal of 4 additional models - B73F and B757 remodelled - MD90 added as equivalence model - Cruise and descent speeds for several turboprops changed - Climb thrust for several a/c changed - Removal of C_{m16} from drag expression
Revision 3.1 Issue 1.0	01.10.98	Released with BADA Revision 3.1 - Descent & cruise speeds for several jet aircraft changed: DC9, BA46, CL60 - Descent, cruise & climb speeds for several turboprops changed: D228, SH36 - Maximum Operating speed for several a/c changed: PA42 - Stalling speed for several a/c changed: DC8, T154 - Removed formula for air density calculation above tropopause - Addition of Appendix D: Solutions for buffeting limit algorithm - Removed Section 3.7.2: Maximum Take-Off Thrust - Description for Cred parameter added

Issue Number	Release Date	Comments
		- Correction of some minor typing errors - Modified PTF File format (Flight Level): Section 6.6 - Cruise CAS schedule for jet & turbo aircraft (Section 4.2)
Revision 3.3 Issue 1.0		Released with BADA Revision 3.3 - Standard atmosphere explanation added - Correction of some typing errors, minor changes in the layout and equations presentation. - Several aircraft types have changed ICAO's designator according to the ICAO doc.8643/27. Aircraft types affected by the RD3 are as follows: A300, ATR, B707, B727, B73A, B73B, B73C, B74A, B74B, B757, B767, B777, CARJ, DC8, DHC8, JSTA, JSTB, P31T, PA28, PA42. That resulted in: modification of the name of the OPF and APF files, addition of new models as synonyms, modification of Synonym.NEW and Synonym.LST files. - B73A, B757, MD80, B73B, F100, B727, CARJ, FA20, FA50, D228, T154 aircraft models have been re-modelled - A319, A321, A306, AT72 models have been added - Climb, cruise and descent speeds changed for several models. - Ground TOL for B73C has been modified. - MD80: Cd0 and Cd2 for IC and TO added, maximum altitude at MTOW, ISA weight gradient on maximum altitude Gw and temperature gradient Gt on maximum altitude have been changed - BA46 maximum altitude at MTOW, ISA weight gradient on maximum altitude Gw have been changed - E145 was added as equivalent of CRJ1 - A478 was added as equivalent of AT72
Revision 3.4 Issue 1.0	June 2002	Released with BADA Revision 3.4 - correction of some typing errors - in chapter 3.5 configuration threshold altitude values replaced with $H_{max,i}$ while the corresponding numbers are listed in chapter 5.6 - Appendix B: a new column is added to the table; providing the information on maximum altitude that an aircraft can reach at MTOW (h_{max}) - FGTN aircraft model added - FGTH aircraft model added - FGTL aircraft model added - FGTR aircraft model removed

Issue Number	Release Date	Comments
		- DC-9 aircraft model re-modelled - D228 cruise and descent speed modified - SH36 cruise and descent speed modified - B738 maximum operational altitude modified - AT72 cruise speed corrected - PA34 minimum mass modified - B734 aircraft model added - B735 aircraft model added - E145 aircraft model added - B737 aircraft model added - AT45 aircraft model added - B762 aircraft model added - B743 aircraft model added - Removal of several existing OPF and APF files due to the change of ICAO aircraft designators according to RD3: A330, A340, BA46, DC9, MD80 - Addition of several new OPF and APF files due to the change of ICAO aircraft designators according to RD3: A333, A343, B461, DC94, MD83 - Addition of new equivalence aircraft types: A332, A342, A345, A346, B461, B462, B463, DC91, DC92, DC93, DC95, MD81, MD82, MD87, MD88, A124, AC80, AC90, AC95, AJET, AMX, AN72, ATLA, B1, B350, B739, B74D, BDOG, BE10, BE40, BE76, BER4, C17, C72R, C77R, C82R, C210, C212, C337, C526, C56X, CRJ7, E135, EUFI, F1, FT2H, F104, G222, GLF5, HAWK, H25A, H25C, IL96, JS1, JS3, JS20, LJ24, M20T, M20P, K35R, N262, P28T, P28B, PA32, PAY4, P68, PA44, SB05, T204, TBM7 - Modification of the value for Maximum bank angles for civil flight during HOLD in BADA.GPF file - Configuration Management of BADA files have been changed; files have been migrated from RCS to Continuus Configuration Management System. That resulted in the modification of the “identification” part of all BADA files given in the header.
Revision 3.5 Issue 1.0	July 2003	Released with BADA Revision 3.5 - correction of some typing errors - B712 aircraft model added - LJ45 aircraft model added - C750 aircraft model added - RJ85 aircraft model added

Issue Number	Release Date	Comments
		- B736 aircraft model added - B753 aircraft model added - A332 aircraft model added - B772 re-modelled - B738 re-modelled - B763 re-modelled - B703 WTC modified - JS41 WTC modified - Addition of new syn. aircraft types: P180, GLEX, C30J, J328, A7, B52, ETAR, F117, L159 - Modification of BADA models for existing synonym aircraft types: C17, GLF3, GLF3, GLF4, GLF5 - SYNONYM_ALL.LST file added.
Revision 3.6 Issue 1.0	July 2004	Released with BADA Revision 3.6 The following models of aircraft added in BADA 3.6: - Dash 8-100: **DH8A** - Boeing MD82: **MD82** - Boeing B767-400: **B764** - Boeing B777-300: **B773** - BAE 146-200: **B462** The following models of aircraft have been re-modelled in BADA 3.6: - Airbus A300B4-203: **A30B** - Airbus A310: **A310** - Airbus A319: **A319** - Airbus A320: **A320** - Airbus A321: **A321** - Airbus A330-301: **A333** - Airbus A340-313: **A343** - Boeing B737-200: **B732** - Boeing B737-300: **B733** - Boeing B747-200: **B742** - Boeing B747-400: **B744** - Boeing B757-200: **B752** Addition of new synonym aircraft types: A3ST, ASTR, B701, C441, GALX, J728, K35A, K35E, L29B, LJ25, LJ60, NIM, PC12, R135, RJ1H, RJ70, P32R, C208, AA5, S76, DC3, BLAS, AEST, EC35, PAY1, PA18, BE55, C170, B461. Correction of syntax errors in BADA files: - Boeing B777-200: **B772** - ATR42-500: **AT45**

Issue Number	Release Date	Comments
Revision 3.7 Issue 1.0	March 2009	Released with BADA Revision 3.7 - Modification of the values for constants g and R in Section 3. - New description of formula 3.1-8 to match its actual use in some models. - Coefficient CVmin, TO is no longer used in climb speed schedule, only in flight envelope determination. - Numbering of several equations changed due to reorganisation of related sections. - Change of descent thrust computation when CTdes,app and CTdes, ld are null in Section 3.7.3. - Clarification of descent fuel flow computation in Section 3.9. - Additional information on climb and descent speed schedules in Section 4. - Update of some Fortran format descriptions in Section 6. - Additional reasons for ROCD discontinuities added in Section 6.6. - Introduction of new PTD file format. - Update of Section 7 to describe the new means of access to the BADA files. - Remodelling of 71 a/c types from BADA 3.6 - more details in [RD8]. - Addition of 12 new a/c models for following a/c types: A346, A388, BE58, C510, CRJ2, CRJ9, DA42, DH8D, E135, E170, E190, EA50. - All synonym aircraft have been re-evaluated and some reassigned – more details in [RD12] reassigned. -
Revision 3.8 Issue 1.0	April 2010	Released with BADA Revision 3.8 - Introduction of new revised atmosphere model and relevant corresponding updates throughout the User Manual document - Harmonisation of acronyms for physical constants with the EEC Technical Report No. 2010-001, February 2010 “Revision of Atmosphere Model in BADA Aircraft Performance Model” - Clarification of descent fuel flow computation in Section 3.9. - Information added on whether some BADA model coefficients may or may not be negative. - Missing information about speed schedule in cruise for piston aircraft added (section 4.2) - Additional clarifications provided on use of altitudes in Section 4. - Additional explanatory note provided on data

Issue Number	Release Date	Comments
		presented in the PTF file. - Correction of error in the solution for buffeting limit algorithm. - Remodelling of 5 a/c types from BADA 3.7: B763, FA50, F900, RJ85, TRIN - Addition of 8 new a/c models: A318, A3ST, A345, B739, B77L, B77W, F2TH, FA7X. - 23 new synonym aircraft added – more details in [RD12]. - Regeneration of all PTF/PTD files
Revision 3.8 Issue 1.1	August 2010	Clarifications only, no impact on BADA implementations: - Overall review of the document to fix formatting and typography problems. - Formula 3.1-19 (approximate value of a constant) removed, formula 3.1-4 added to define $T_{ISA,trop}$, and some formulas reordered in section 3.1
Revision 3.9 Issue 1.0	April 2011	Released with BADA Revision 3.9: - Minor updates in the document - Clarification about speed calculation in Chapter 4.2. Cruise - Remodelling of 4 a/c types from BADA 3.8: A320, BE58, DA42, E135 - Addition of 6 new a/c models: AT72, AT75, C56X, E50P, E55P,TBM7 - 17 new synonym aircraft added and 13 existing synonyms have been revised
Revision 3.10 Issue 1.0	April 2012	Released with BADA Revision 3.10: - Corrected Fortran specification of the PTF file to match actual release files (it would miss the first digit of descent fuel flow in some cases) - Clarification about the impact of speed envelope on speed calculation in Chapters 4.1, 4.2 and 4.3 - Slight change in the description of the buffeting limit algorithm to mention that the discriminant is not “always” but “usually” negative - Addition of 10 new a/c models: A342, B463, B748, B788, C172, C182, P180, RJ1H, SR22, TBM8 - Full remodelling of 9 a/c types: A343, B462, C560, DH8D, F50, PA34, RJ85, SF34, TBM7

Issue Number	Release Date	Comments
		- Partial update of 16 a/c types: A3ST, A318, A345, A388, B722, B735, B739, B743, B763, B772, B77L, B77W, BE20, C56X, E190, F100 - Addition of 61 new synonym aircraft and revision of 18 existing synonyms
Revision 3.11 Issue 1.0	May 2013	Released with BADA Revision 3.11: - Addition of 23 new a/c models: A124, A140, A148, AN24, AN28, AN30, AN32, AN38, C25A, C25B, C25C, C525, C680, H25B, IL76, IL86, IL96, P28T, PA44, PA46, T204, YK40, YK42 - Partial update of 4 a/c types: A318, C130, DC10, F70 - Addition of 1 new synonym aircraft and revision of 10 existing synonyms Special thanks to Zlata Belotic (University of Belgrade) for her contribution to this release.

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1. INTRODUCTION

1.1. IDENTIFICATION

This document is the User Manual for the Base of Aircraft Data (BADA) Revision 3.11. This manual replaces the previous User Manual for BADA Revision 3.10 [RD1].

1.2. PURPOSE

BADA is a collection of ASCII files which specifies operation performance parameters, airline procedure parameters and performance summary tables for 405 aircraft types. This information is designed for use in trajectory simulation and prediction algorithms within the domain of Air Traffic Management (ATM). All files are maintained within a configuration management system at the EUROCONTROL Validation Infrastructure Centre of Expertise located at the EUROCONTROL Experimental Centre (EEC) in Brétigny-sur-Orge, France.

This document describes the mathematical models on which the data is based and specifies the format of the files which contain the data. In addition, this document describes how the files can be remotely accessed.

1.3. DOCUMENT ORGANISATION

This document consists of seven sections including Section 1, the Introduction. A list of referenced documents along with a glossary of acronyms and symbols are included in this section.

Section 2: Revision Summary, summarises the differences between BADA Revision 3.11 and the previous BADA Revision 3.10.

Section 3: Operation Performance Models, defines the set of equations, which are used to parameterise aircraft performance. This includes models of aerodynamic drag, engine thrust, and fuel consumption. An atmosphere model is also provided.

Section 4: Airline Procedure Models, defines the set of parameters which is used to characterise standard airline speed procedures for climb, cruise, and descent.

Section 5: Global Aircraft Parameters, defines the set of global aircraft parameters that are valid for all, or a group of, aircraft.

Section 6: File Structure, describes the files in which the BADA aircraft parameters are maintained. Six types of files are identified:

- Synonym Files listing the supported aircraft types;
- Operations Performance Files (OPF) containing the performance parameters for a specific aircraft type;
- Airline Procedures Files (APF) containing speed procedure parameters for a specific aircraft type;
- Performance Table Files (PTF) containing summary performance tables of true airspeed, climb/descent rates and fuel consumption at various flight levels for a specific aircraft type;
- Performance Table Data (PTD) containing detailed performance data at various flight levels for a specific aircraft type;

- Global Parameters File (GPF) containing parameters that are valid for all aircraft or a group of aircraft, for instance all turboprops or all military aircraft.

Section 7: Remote File Access to BADA, provides instructions on how to remotely access BADA files from the EUROCONTROL computing facilities over the Internet.

Two appendices are also provided with this document. **Appendix A** provides a list of the aircraft types supported by BADA Revision 3.11 and **Appendix B** gives solutions for a buffeting limit algorithm.

1.4. REFERENCED DOCUMENTS

RD1 User Manual for the Base of Aircraft Data (BADA) Revision 3.10; EEC Technical/Scientific Report No. 12/04/10-45, April 2012.

RD2 Aircraft Type Designators, ICAO Document 8643/40, 2012 edition, http://www.icao.int/publications/DOC8643/

RD3 Aircraft Modelling Standards for Future ATC Systems; EUROCONTROL Division E1 Document No. 872003, July 1987.

RD4 Manual of the ICAO Standard Atmosphere; ICAO Document No. 7488, 2nd Edition, 1964.

RD5 BADA Product Management Document; EEC Technical Report No. 2009-008, April 2009.

RD6 Base of Aircraft Data (BADA) Aircraft Performance Modelling Manual: EEC Technical Report No. 2009-009, April 2009.

RD7 Memo on the Calculation of Energy Share Factor; EEC/FAS/BYR/95/50; 22 November 1995.

RD8 Revision Summary Document for the Base of Aircraft Data (BADA) Revision 3.11; EEC Technical/Scientific Report No. 13/04/16-02; May 2013.

RD9 Aircraft Performance Summary Tables for the Base of Aircraft Data (BADA) Revision 3.11; EEC Technical/Scientific Report No. 13/04/16-03; May 2013.

RD10 Aircraft Type Designators, ICAO Document 8643, Versions 24-40.

RD11 BADA Support Application – User Guide, revision 1.1, August 2009.

RD12 Synonym Aircraft Report for the Base of Aircraft Data (BADA) - Revision 3.11: EEC Technical/Scientific Report No. 13/04/16-05, May 2013.

RD13 Model Accuracy Summary Report for the Base of Aircraft Data (BADA) - Revision 3.11: EEC Technical/Scientific Report No. 13/04/16-04, May 2013.

RD14 Revision of Atmosphere Model in BADA Aircraft Performance Model: EEC Technical Report No. 2010-001, February 2010.

RD15 Mathematical Handbook; M.R. Spiegel; 1968; McGraw-Hill book company.

1.5. GLOSSARY OF ACRONYMS

AGL	**A**bove **G**round **L**evel
APF	**A**irlines **P**rocedures **F**ile
ASCII	**A**merican **S**tandard **C**ode for the **I**nterchange of **I**nformation
ATM	**A**ir **T**raffic **M**anagement
BADA	**B**ase of **A**ircraft **Da**ta
CAS	**C**alibrated **Air**speed
EEC	**E**UROCONTROL **E**xperimental **C**entre
ESF	**E**nergy **S**hare **F**actor
ICAO	**I**nternational **C**ivil **A**viation **O**rganisation
ISA	**I**nternational **S**tandard **A**tmosphere
MLW	**M**aximum **L**anding **W**eight
MSL	**M**ean **S**ea **L**evel
MTOW	**M**aximum **T**ake-**o**ff **W**eight
OPF	**O**perations **P**erformance **F**ile
PTD	**P**erformance **T**able **D**ata
PTF	**P**erformance **T**able **F**ile
RCS	**R**evision **C**ontrol **S**ystem
ROCD	**R**ate of **C**limb or **D**escent
TAS	**T**rue **Air**speed
TEM	**T**otal-**E**nergy **M**odel

1.6. GLOSSARY OF SYMBOLS

A list of the symbols used in equations throughout this document is given below along with a description. Where appropriate, the engineering units typically associated with the symbol are also given.

a	speed of sound	[m/s]
d	distance	[nautical miles]
f	fuel flow	[kg/min]
g_0	gravitational acceleration	[m/s^2]
$\frac{dh}{dt}$	vertical speed	[m/s] or [ft/min]
h	geodetic altitude	[metres] or [ft]
H	geopotential altitude	[metres] or [ft]
H_p	geopotential pressure altitude	[metres] or [ft]
C	general coefficient	
D	drag force	[Newtons]
m	aircraft mass	[tonnes] or [kg]
M	Mach number	[-]
p	Actual pressure	[Pa]
p_0	Standard pressure at MSL	[Pa]
R	real gas constant for air	[$m^2/(K \cdot s^2)$]
ROCD	Rate of Climb or Descent	[m/s] or [ft/min]
S	reference wing surface area	[m^2]
T	temperature	[Kelvin]
Thr	thrust	[N]
V	speed	[m/s] or [knots]
ΔT	temperature difference	[Kelvin]
W	weight	[N]
η	thrust specific fuel flow	[kg/(min·kN)]
ρ	air density	[kg/m^3]

2. REVISION SUMMARY

This section summarises the aircraft types that are supported in BADA Revision 3.11 along with the updates that have been made from the previous release, BADA Revision 3.10.

2.1. SUPPORTED AIRCRAFT

BADA Revision 3.11 provides operations and procedures data for a total of 405 aircraft types. For 150 of these aircraft types, data is provided directly in files. These aircraft types are referred to as being directly supported and referred to as aircraft original models. The way they have been identified is described in [RD6]. For the other 255 aircraft types, the data is specified to be the same as one of the directly supported 150 aircraft types. These aircraft types have been identified as being 'equivalent' to original aircraft models. They are referred to as synonym aircraft. More details on the way they have been identified are given in [RD12].

With three exceptions, each supported aircraft type is identified by a 4-character designation code assigned by the International Civil Aviation Organisation (ICAO) [RD2]. The exceptions are the models representing generic military fighters, which use the designators: FGTH, FGTL, FGTN.

The list of aircraft types supported by BADA Revision 3.11 is given in Appendix A. In this Appendix the supported aircraft types are listed alphabetically by their designation code. For each aircraft type, the aircraft name and type of BADA support (either original or synonym) is specified. Also, for each synonym aircraft, which is supported through equivalence, the corresponding equivalent aircraft type is specified.

2.2. UPDATES FOR BADA REVISION 3.11

Updates made to BADA Revision 3.11 from the previous Revision 3.10 are listed below:

(a) Updates of existing documentation.

(b) Addition of 23 new aircraft models.

(c) Partial update of 4 aircraft models.

(d) Addition of new synonym aircraft and revision of existing ones.

(e) Implementation of new ICAO aircraft designators according to the ICAO Doc. 8643 [RD2].

A more complete overview of all changes can be found in [RD8].

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3. OPERATIONS PERFORMANCE MODEL

This section defines the various equations and coefficients used by the BADA operations performance model.

The first two subsections describe the equations for atmospheric properties and the Total-Energy Model (TEM) equations respectively.

The remaining eight subsections define the aircraft model in terms of the eight categories listed below:

- aircraft type,
- mass,
- flight envelope,
- aerodynamics,
- engine thrust,
- reduced power,
- fuel consumption,
- ground movement.

3.1. ATMOSPHERE MODEL

This section provides expressions for the atmospheric properties (pressure, temperature, density and speed of sound) as a function of altitude which are required for calculation of aircraft performances and movements[1]. Conversions from CAS to TAS and Mach number also require the determination of several atmospheric properties as a function of altitude.

The most important equations for atmospheric properties used by BADA and CAS/TAS conversion are summarised in this chapter, while other expressions and more details are provided in [RD14].

3.1.1. Definitions

Mean Sea Level (MSL) Standard atmosphere conditions are those that occur in the International Standard Atmosphere (ISA) at the point where the geopotential pressure altitude H_p[2] is zero. They are denoted as T_0, p_0, ρ_0 and a_0 with the values listed below:

Standard atmospheric temperature at MSL:	T_0	=	288.15	[K]
Standard atmospheric pressure at MSL:	p_0	=	101325	[Pa]
Standard atmospheric density at MSL:	ρ_0	=	1.225	[kg/m^3]
Speed of sound:	a_0	=	340.294	[m/s]

[1] These equations are based on the International Standard Atmosphere (ISA) [RD4].

[2] Geopotential pressure altitude H_p is the geopotential altitude H that occurs in the ISA atmospheric conditions [RD14].

Mean Sea Level (MSL) atmosphere conditions are those that occur in a non-ISA atmosphere. They are identified by the sub-index MSL and differ from (T_0, p_0, ρ_0, a_0) in non-ISA conditions.

Non-ISA atmospheres are those that follow the same hypotheses as the ISA atmosphere but differ from it in that one or both of the following parameters is not zero:

1. **ΔT**. Temperature differential at MSL. It is the difference in atmospheric temperature at MSL between a given non-standard atmosphere and ISA.

2. **Δp**. Pressure differential at MSL. It is the difference in atmospheric pressure at MSL between a given non-standard atmosphere and ISA.

The values of these two parameters uniquely identify any non-ISA atmosphere. Thus, a non-ISA atmosphere provides expressions for the atmospheric pressure, temperature and density as functions of the geopotential altitude H[3] and its two differentials. [RD14] provides more details on the corresponding analytical expressions.

3.1.2. Expressions

The relationships linking the atmospheric pressure p, temperature T, geopotential pressure altitude H_p and geopotential altitude H for any ISA[4] and non-ISA atmosphere are provided below.

Physical constants which are used throughout this chapter are listed below:

Adiabatic index of air:	κ	=	1.4	
Real gas constant for air:	R	=	287.05287	$[m^2/(K \cdot s^2)]$
Gravitational acceleration:	g_0	=	9.80665	$[m/s^2]$
ISA temperature gradient with altitude below the tropopause:	$\beta_{T,<}$	=	- 0.0065	[K/m]

Note that subindex < denotes values below and at the tropopause and subindex > denotes values above the tropopause (as defined by 3.1-11).

Standard Mean Sea Level (subindex $H_p = 0$)

The temperature differential ΔT sets the value of the real temperature T in non-standard atmospheres.

$$H_{p,Hp=0} = 0 \qquad (3.1\text{-}1)$$

[3] Geopotential altitude H is that which under the standard constant gravitational field provides the same differential work performed by the standard acceleration of free fall when displacing the unit of mass a distance dH along the line of force, as that performed by the geopotential acceleration when displacing the unit of mass a geodetic distance dh [RD14].

[4] By replacing ΔT and Δp parameters with zeros the expressions are made applicable to the case of the standard atmosphere.

$$p_{Hp=0} = p_0 \quad (3.1\text{-}2)$$

$$T_{ISA,Hp=0} = T_0 \quad (3.1\text{-}3)$$

$$T_{Hp=0} = T_0 + \Delta T \quad (3.1\text{-}4)$$

$$H_{Hp=0} = \frac{1}{\beta_{T,<}}\left[T_0 - T_{ISA,MSL} + \Delta T \cdot Ln\left(\frac{T_0}{T_{ISA,MSL}}\right)\right] \quad (3.1\text{-}5)$$

where T_{ISA} is the standard atmospheric temperature that occurs in the ISA atmosphere. It is a function of the geopotential pressure altitude H_p.

Mean Sea Level (subindex MSL)

The pressure differential Δp sets the value of the atmospheric pressure p.

$$H_{MSL}=0 \quad (3.1\text{-}6)^5$$

$$p_{MSL} = p_0 + \Delta p \quad (3.1\text{-}7)$$

$$H_{p,MSL} = \frac{T_0}{\beta_{T,<}}\left[\left(\frac{p_{MSL}}{p_0}\right)^{-\frac{\beta_{T,<}R}{g_0}} - 1\right] \quad (3.1\text{-}8)$$

$$T_{ISA,MSL} = T_0 + \beta_{T,<} H_{p,MSL} \quad (3.1\text{-}9)$$

$$T_{MSL} = T_0 + \Delta T + \beta_{T,<} H_{p,MSL} \quad (3.1\text{-}10)$$

Tropopause

Tropopause is the separation between two different layers: the troposphere, which stands below it, and the stratosphere, which is placed above. Its altitude $H_{p,trop}$ is constant when expressed in terms of geopotential pressure altitude:

$$H_{p,trop} = 11000 \text{ [m]} \quad (3.1\text{-}11)$$

[5] In order to simplify the expressions, this document assumes that the geopotential altitude at mean sea level is always zero.

a) Determination of Temperature

$$T = f(H_p, \Delta T) \tag{3.1-12}$$

$$T_< = T_0 + \Delta T + \beta_{T,<} H_{p,<} \tag{3.1-13}$$

$$T_{ISA,trop} = T_0 + \beta_{T,<} H_{p,trop} \tag{3.1-14}$$

$$T_{trop} = T_0 + \Delta T + \beta_{T,<} H_{p,trop} \tag{3.1-15}$$

$$T_> = T_{trop} \tag{3.1-16}$$

b) Determination of Air Pressure

$$p = f(T, \Delta T) \tag{3.1-17}$$

$$p_< = p_0 \left(\frac{T_< - \Delta T}{T_0} \right)^{-\frac{g_0}{\beta_{T,<} R}} \tag{3.1-18}$$

$$p_{trop} = p_0 \left(\frac{T_{trop} - \Delta T}{T_0} \right)^{-\frac{g_0}{\beta_{T,<} R}} \tag{3.1-19}$$

$T_> = T_{trop}$, so $p_>$ does not directly depend on temperature $T_>$. For altitudes above the tropopause, the following formula should be used:

$$p_> = p_{trop} \exp\left[-\frac{g_0}{R\, T_{ISA,trop}} \left(H_{p,>} - H_{p,trop} \right) \right] \tag{3.1-20}$$

where altitudes $H_{p,>}$ and $H_{p,trop}$ are expressed in metres.

c) Determination of Air Density

The air density, ρ, in kg/m^3, is calculated from the pressure p and the temperature T at altitude using the perfect gas law:

$$\rho = \frac{p}{R\, T} \tag{3.1-21}$$

d) Determination of Speed of Sound

The speed of sound, a, is the speed at which the pressure waves travel through a fluid and it is given by the expression:

$$a = \sqrt{\kappa R T} \tag{3.1-22}$$

e) CAS/TAS Conversion

The true airspeed, V_{TAS}, is calculated as a function of the calibrated air speed, V_{CAS}, as follows:

$$V_{TAS} = \left[\frac{2}{\mu}\frac{p}{\rho}\left\{\left(1+\frac{p_0}{p}\left[\left(1+\frac{\mu}{2}\frac{\rho_0}{p_0}V_{CAS}^{\ 2}\right)^{\frac{1}{\mu}}-1\right]\right)^{\mu}-1\right\}\right]^{\frac{1}{2}} \tag{3.1-23}$$

Similarly, V_{CAS} is calculated as a function of V_{TAS} as follows:

$$V_{CAS} = \left[\frac{2}{\mu}\frac{p_0}{\rho_0}\left\{\left(1+\frac{p}{p_0}\left[\left(1+\frac{\mu}{2}\frac{\rho}{p}V_{TAS}^{\ 2}\right)^{\frac{1}{\mu}}-1\right]\right)^{\mu}-1\right\}\right]^{\frac{1}{2}} \tag{3.1-24}$$

where symbols not previously defined are explained below:

$$\mu = \frac{\kappa - 1}{\kappa} \qquad \left(\mu = \frac{1}{3.5} \text{ if } \kappa = 1.4\right) \tag{3.1-25}$$

Note that for these conversion formulas above, the speeds V_{TAS} and V_{CAS} must be specified in m/s.

f) Mach/TAS conversion

The true airspeed, V_{TAS} [m/s], is calculated as a function of the Mach number, M, as follows:

$$V_{TAS} = M \times \sqrt{\kappa R T} \tag{3.1-26}$$

g) Mach/CAS transition altitude

The transition altitude (also called crossover altitude), $H_{p,trans}$ [ft], between a given CAS, V_{CAS} [m/s], and a Mach number, M, is defined to be the geopotential pressure altitude at which V_{CAS} and M represent the same TAS value, and can be calculated as follows:

$$H_{p,trans} = \left(\frac{1000}{0.3048 \cdot 6.5} \right) \cdot \left[T_0 \cdot \left(1 - \theta_{trans} \right) \right] \quad (3.1\text{-}27)$$

where θ_{trans} is the temperature ratio at the transition altitude,

$$\theta_{trans} = (\delta_{trans})^{-\frac{\beta_{T,<} R}{g_0}} \quad (3.1\text{-}28)$$

where δ_{trans} is the pressure ratio at the transition altitude,

$$\delta_{trans} = \frac{\left[1 + \left(\frac{\kappa - 1}{2} \right) \left(\frac{V_{CAS}}{a_0} \right)^2 \right]^{\frac{\kappa}{\kappa - 1}} - 1}{\left[1 + \frac{\kappa - 1}{2} M^2 \right]^{\frac{\kappa}{\kappa - 1}} - 1} \quad (3.1\text{-}29)$$

3.2. TOTAL-ENERGY MODEL

The Total-Energy Model equates the rate of work done by forces acting on the aircraft to the rate of increase in potential and kinetic energy, that is:

$$(Thr - D) \cdot V_{TAS} = mg_0 \frac{dh}{dt} + mV_{TAS} \frac{dV_{TAS}}{dt} \qquad (3.2\text{-}1)$$

The symbols are defined below with metric units specified:

Thr	-	thrust acting parallel to the aircraft velocity vector	[Newtons]
D	-	aerodynamic drag	[Newtons]
m	-	aircraft mass	[kilograms]
h	-	geodetic altitude	[m]
g_0	-	gravitational acceleration	[9.80665 m/s^2]
V_{TAS}	-	true airspeed	[m/s]
$\frac{d}{dt}$	-	time derivative	[s^{-1}]

Note that true airspeed is often calculated in knots and altitude calculated in feet thus requiring the appropriate conversion factors.

Without considering the use of devices such as spoilers, leading-edge slats or trailing-edge flaps, there are two independent control inputs available for affecting the aircraft trajectory in the vertical plane. These are the throttle and the elevator.

These inputs allow any two of the three variables of thrust, speed, or rate of climb or descent (ROCD) to be controlled. The other variable is then determined by equation 3.2-1. The three resulting control possibilities are elaborated on below.

(a) Speed and Throttle Controlled - Calculation of Rate of Climb or Descent

Assuming that velocity and thrust are independently controlled, then equation 3.2-1 is used to calculate the resulting rate of climb or descent (ROCD). This is a fairly common case for climbs and descents in which the throttle is set to some fixed position (maximum climb thrust or idle for descent) and the speed is maintained at some constant value of calibrated airspeed (CAS) or Mach number.

(b) ROCD and Throttle Controlled - Calculation of Speed

Assuming that the ROCD and thrust are independently controlled, then equation 3.2-1 is used to calculate the resulting speed.

(c) Speed and ROCD Controlled - Calculation of Thrust

Assuming that both ROCD and speed are controlled, then equation 3.2-1 can be used to calculate the necessary thrust. This thrust must be within the available limits for the desired ROCD and speed to be maintained.

Case (a), above, is the most common such that equation 3.2-1 is most often used to calculate the rate of climb or descent. To facilitate this calculation, equation 3.2-1 can be rearranged as follows:

$$(Thr - D) \cdot V_{TAS} = mg_0 \frac{dh}{dt} + m V_{TAS} \left(\frac{dV_{TAS}}{dh} \right) \left(\frac{dh}{dt} \right) \quad (3.2\text{-}2)$$

Isolating the vertical speed on the left hand side gives:

$$\frac{dh}{dt} = \frac{(Thr - D) \cdot V_{TAS}}{mg_0} \left[1 + \left(\frac{V_{TAS}}{g_0} \right) \left(\frac{dV_{TAS}}{dh} \right) \right]^{-1} \quad (3.2\text{-}3)$$

Vertical speed is defined as the variation with time of the aircraft geodetic altitude h. The assumption of a standard constant gravity field derives in identical geodetic and geopotential altitudes H [RD14].

The ROCD is defined as the variation with time of the aircraft geopotential pressure altitude H_p. It is the preferred way of presenting the performances of an aircraft as it eliminates possible variations caused by the atmospheric conditions:

$$ROCD = \frac{dH_p}{dt} = \frac{T - \Delta T}{T} \frac{(Thr - D) \cdot V_{TAS}}{mg_0} \left[1 + \left(\frac{V_{TAS}}{g_0} \right) \left(\frac{dV_{TAS}}{dh} \right) \right]^{-1} \quad (3.2\text{-}4)$$

where:

T	-	atmosphere temperature [K];
ΔT	-	temperature differential [K].

It has been shown by Renteux [RD3] that the last term can be replaced by an energy share factor as a function of Mach number, f{M}:

$$f\{M\} = \left[1 + \left(\frac{V_{TAS}}{g_0} \right) \cdot \left(\frac{dV_{TAS}}{dh} \right) \right]^{-1} \quad (3.2\text{-}5)$$

This leads to:

$$\frac{dh}{dt} = \left[\frac{(Thr - D) \cdot V_{TAS}}{mg_0} \right] f\{M\} \quad (3.2\text{-}6)$$

$$ROCD = \frac{dH_P}{dt} = \frac{T - \Delta T}{T} \left[\frac{(Thr - D) \cdot V_{TAS}}{mg_0} \right] f\{M\} \quad (3.2\text{-}7)$$

This energy share factor f{M} specifies how much of the available power is allocated to climb as opposed to acceleration while following a selected speed profile during climb.

For several common flight conditions, equation 3.2-5 can be rewritten as is done below. A more comprehensive description of this process can be found in [RD7]:

(a) Constant Mach number in stratosphere (i.e. above tropopause)

$$f\{M\} = 1.0 \tag{3.2-8}$$

Note that above the tropopause the air temperature and the speed of sound are constant. Maintaining a constant Mach number therefore requires no acceleration and all available power can be allocated to a change in altitude.

(b) Constant Mach number below tropopause:

$$f\{M\} = \left[1 + \frac{\kappa R \beta_{T,<}}{2 g_0} M^2 \frac{T - \Delta T}{T}\right]^{-1} \tag{3.2-9}$$

In this case, for a typical Mach number of 0.8 the energy share factor allocated to climb is 1.09.

This number is greater than 1 because below the tropopause, the temperature and thus, speed of sound decreases with altitude. Maintaining a constant Mach number during climb thus means that the true airspeed decreases with altitude. Consequently, the rate of climb benefits from not only all the available power but also a transfer of kinetic energy to potential energy.

(c) Constant Calibrated Airspeed (CAS) below tropopause

$$f\{M\} = \left\{1 + \frac{\kappa R \beta_{T,<}}{2 g_0} M^2 \frac{T - \Delta T}{T} + \left(1 + \frac{\kappa - 1}{2} M^2\right)^{\frac{-1}{\kappa - 1}} \left\{\left(1 + \frac{\kappa - 1}{2} M^2\right)^{\frac{\kappa}{\kappa - 1}} - 1\right\}\right\}^{-1} \tag{3.2-10}$$

In this case the energy share factor is less than one. A Mach number of 0.6 for example yields an energy share factor of 0.85.

This number is less than 1 because as density decreases with altitude, maintaining a constant CAS during climb requires maintaining a continual increase in true airspeed. Thus, some of the available power needs to be allocated to acceleration leaving the remainder for climb.

(d) Constant Calibrated Airspeed (CAS) above tropopause.

$$f\{M\} = \left\{1 + \left(1 + \frac{\kappa - 1}{2} M^2\right)^{\frac{-1}{\kappa - 1}} \left\{\left(1 + \frac{\kappa - 1}{2} M^2\right)^{\frac{\kappa}{\kappa - 1}} - 1\right\}\right\}^{-1} \tag{3.2-11}$$

This formula is identical to (3.2-10), except that β_T is now null since we are above the tropopause.

The energy share factors given above apply equally well to descent as to climb. The difference being that the available power is negative for descent.

In cases where neither constant Mach number nor constant CAS is maintained, the following energy share factors are used:

- acceleration in climb: f{M} = 0.3
- deceleration in descent: f{M} = 0.3
- deceleration in climb: f{M} = 1.7
- acceleration in descent: f{M} = 1.7

Note that, for the cases of acceleration in climb or deceleration in descent, the majority of the available power is devoted to a change in speed.

For the cases of deceleration in climb or acceleration in descent, the energy share factor is greater than 1 since the change of altitude benefits from a transfer of kinetic energy.

3.3. AIRCRAFT TYPE

Three values are specified for aircraft type, these being the number of engines, n_{eng}, the engine type and the wake category.

The engine type can be one of three values:

- Jet
- Turboprop
- Piston

The wake category can also be one of four values:

- J : jumbo
- H : heavy
- M: medium
- L : light

Note that ICAO associates a wake category with each aircraft type designator [RD2].

3.4. MASS

Four mass values are specified for each aircraft in tonnes:

m_{min} - minimum mass
m_{max} - maximum mass
m_{ref} - reference mass
m_{pyld} - maximum payload mass

Note that the specified mass limits are taken from aircraft performance reference data which is available in the BADA library. In function of specific aircraft certified limitations, a particular aircraft version of a given aircraft type (model) may have different limits. More details on the way the mass limits are selected in BADA are provided in [RD6].

Aircraft operating speeds vary with the aircraft mass. This variation is calculated according to the formula below:

$$V = V_{ref} \times \sqrt{\frac{m}{m_{ref}}} \qquad (3.4\text{-}1)$$

In this formula, the aircraft reference speed V_{ref} is given for the reference mass m_{ref}. The speed at another mass, m, is then calculated as V.

An example of an aircraft speed which can be calculated via this formula is the stall speed, V_{stall}.

3.5. FLIGHT ENVELOPE

(a) Maximum Speed and Altitude

The maximum speed and altitude for an aircraft are expressed in terms of the following six parameters:

V_{MO}	-	maximum operating speed (CAS) [kt]
M_{MO}	-	maximum operational Mach number
h_{MO}	-	maximum operating altitude [ft] above standard MSL
h_{max}	-	maximum altitude [ft] above standard MSL at MTOW under ISA conditions (allowing about 300 ft/min of residual rate of climb)
G_w	-	mass gradient on h_{max} [ft/kg]
G_t	-	temperature gradient on h_{max} [ft/K]

The maximum altitude for any given mass is:

$$h_{max/act} = MIN\left[h_{MO}, h_{max} + G_t \times \left(\Delta T - C_{Tc,4}\right) + G_w \times \left(m_{max} - m_{act}\right)\right] \quad (3.5\text{-}1)$$

where: ΔT is the temperature deviation from ISA [K]
m_{act} is the actual aircraft mass [kg]

with: $G_w \geq 0$

$G_t \leq 0$

if $(\Delta T - C_{Tc,4}) < 0$, then: $(\Delta T - C_{Tc,4}) = 0$

Formula 3.5-1 should not be executed when the h_{max} value in the OPF file is set to 0 (zero). In that case the maximum altitude is always h_{MO}.

Note that the given speed and altitude limits are taken from available reference data: depending upon specific certifications, a particular aircraft of a given type may present different limits.

(b) Minimum Speed

The minimum speed for the aircraft is in function of aircraft stall speed and specified as follows:

$$V_{min} = C_{Vmin,TO} \times V_{stall} \quad \text{if in take-off} \quad (3.5\text{-}2)$$

$$V_{min} = C_{Vmin} \times V_{stall} \quad \text{otherwise} \quad (3.5\text{-}3)$$

Note: See Section 3.6.2 for minimum speed at high altitude for jet aircraft and Section 5.7 for the values of the minimum speed coefficients.

Here the speeds are specified in terms of CAS. The stall speed depends upon the configuration.

Specifically, five different configurations are specified with a stall speed, $(V_{stall})_i$, and configuration threshold altitude, $H_{max,i}$, given for each:

TO - take-off configuration $(V_{stall})_{TO}$
(in climb up to $H_{max,TO}$ AGL)

IC - initial climb configuration $(V_{stall})_{IC}$
(in climb between $H_{max,TO}$ and $H_{max,IC}$ AGL)

CR - cruise (clean) configuration $(V_{stall})_{CR}$
(in climb above $H_{max,IC}$ AGL,
in descent above $H_{max,AP}$ AGL,
in descent below $H_{max,AP}$ AGL when
$V \geq V_{min,cruise} + 10$ kt)

AP - approach configuration $(V_{stall})_{AP}$
(in descent between $H_{max,AP}$ AGL and $H_{max,LD}$ AGL when
$V < V_{min,cruise} + 10$ kt,
in descent below $H_{max,LD}$ AGL when
$V_{min,cruise} + 10 \text{ kt} > V \geq V_{min,approach} + 10$ kt)

LD - landing configuration $(V_{stall})_{LD}$
(in descent below $H_{max,LD}$ AGL when
$V < V_{min,approach} + 10$ kt)

The threshold altitudes are expressed in terms of geopotential pressure altitude. However, when aircraft operations close to the ground are considered, one has to account for airport/runway elevation[6]. The pressure altitude thresholds provided above correspond to geopotential pressure altitude Above Ground Level (AGL).

The stall speeds correspond to a minimum stall speed and not a 1-g stall speed. Also, the BADA model assumes that for any aircraft these stall speeds have the following relationship:

$$(V_{stall})_{CR} \geq (V_{stall})_{IC} \geq (V_{stall})_{TO} \geq (V_{stall})_{AP} \geq (V_{stall})_{LD}$$

The configuration specific values are listed in Section 5.6. The speeds V used during the descent, approach and landing phases are defined in Section 4.3.

[6] Measured from Mean Sea Level (MSL).

3.6. AERODYNAMICS

3.6.1. Aerodynamic Drag

The lift coefficient, C_L, is determined assuming that the flight path angle is zero. However, a correction for a bank angle ϕ is made.

$$C_L = \frac{2 \cdot m \cdot g_0}{\rho \cdot V_{TAS}^{\ 2} \cdot S \cdot \cos\phi} \qquad (3.6\text{-}1)$$

Under nominal conditions, the drag coefficient, C_D is specified as a function of the lift coefficient C_L as follows:

$$C_D = C_{D0,CR} + C_{D2,CR} \times (C_L)^2 \qquad (3.6\text{-}2)$$

Formula 3.6-2 is valid for all situations except for the approach and landing where other drag coefficients are to be used.

In the approach configuration (as defined in Section 3.5) a different flap setting is used, and formula 3.6-3 should be applied:

$$C_D = C_{D0,AP} + C_{D2,AP} \times (C_L)^2 \qquad (3.6\text{-}3)$$

In the landing configuration (as defined in Section 3.5) a different flap setting is used, and formula 3.6-4 should be applied:

$$C_D = C_{D0,LDG} + C_{D0,\Delta LDG} + C_{D2,LDG} \times (C_L)^2 \qquad (3.6\text{-}4)$$

The value of $C_{D0,\Delta LDG}$ represents drag increase due to the landing gear. The values of $C_{D0,LD}$ in the OPF files were all determined for the landing flap setting mentioned in the OPF file.

The drag force [Newtons] is then determined from the drag coefficient in the standard manner:

$$D = \frac{C_D \cdot \rho \cdot V_{TAS}^{\ 2} \cdot S}{2} \qquad (3.6\text{-}5)$$

Where:

ρ is the air density [kg/m^3]

S is the wing reference area [m^2]

V_{TAS} is the true airspeed [m/s].

Note that the air density is a function of altitude as described in Section 3.1.

The above equations thus result in eight coefficients for the specification of drag:

S

$C_{D0,CR}$ $C_{D2,CR}$

$C_{D0,AP}$ $C_{D2,AP}$

$C_{D0,LD}$ $C_{D2,LD}$

$C_{D0,\Delta LDG}$

In case the $C_{D0,AP}$, $C_{D2,AP}$, $C_{D0,LD}$, $C_{D2,LD}$ and $C_{D0,\Delta LDG}$ coefficients (referred to as "non-clean" data in this document) are set to 0 (zero) in the OPF file, expression 3.6-2 will be used in all cases.

3.6.2. Low Speed Buffeting Limit (jet aircraft only)

For jet aircraft a low speed buffeting limit has been introduced. This buffeting limit is expressed as a Mach number and can be determined using the following equation:

$$k \times M^3 - C_{Lbo\,(M=0)} \times M^2 + \frac{W}{S \cdot p \cdot 0.583} = 0 \qquad (3.6\text{-}6)$$

where:

k is lift coefficient gradient

$C_{Lbo\,(M=0)}$ is initial buffet onset lift coefficient for M=0

p is actual pressure [Pa]

M is Mach number

S is the wing reference area [m^2]

W is aircraft weight [N]

Note that the factor of 0.583 gives a 0.2 g margin.

The k and $C_{Lbo\,(M=0)}$ parameters have been determined for nearly all jet aircraft in BADA Revision 3.11. If the k and $C_{Lbo\,(M=0)}$ parameters in the OPF file are set to 0 (zero), the minimum speed is given by expressions 3.5-2 and 3.5-3. Otherwise, the solution for M in Formula 3.6-6 can be obtained using the method given in Appendix B. The buffeting limit should be applied as a minimum speed in the following way:

- If ($H_p \geq 15{,}000$ ft) then: $V_{min} = MAX(V_{min,stall}, M_b)$

- If ($H_p < 15{,}000$ ft) then: $V_{min} = V_{min,stall}$

where: H_p is the geopotential pressure altitude

M_b is the lowest positive solution of expression 3.6-6

$V_{min,stall}$ is given by expressions 3.5-2 and 3.5-3

Note that the units of the two values $V_{min,stall}$ and M_b inside the MAX() expression should be the same.

3.7. ENGINE THRUST

The BADA model provides coefficients that allow the calculation of the following thrust levels:

- maximum climb and take-off,
- maximum cruise,
- descent.

The thrust is calculated in Newtons and includes the contribution from all engines. The subsections below provide the equations for each of the thrust conditions.

3.7.1. Maximum Climb and Take-Off Thrust

The maximum climb thrust at standard atmosphere conditions, $(Thr_{max\ climb})_{ISA}$, is calculated in Newtons as a function of the following information:

- engine type: either Jet, Turboprop or Piston;
- geopotential pressure altitude, H_p [ft];
- true airspeed, V_{TAS} [kt];
- temperature deviation from standard atmosphere, ΔT [K].

The equations corresponding to the three engine types are given below.

Jet:
$$(Thr_{max\ climb})_{ISA} = C_{Tc,1} \times \left(1 - \frac{H_p}{C_{Tc,2}} + C_{Tc,3} \times H_p^{\ 2}\right) \quad (3.7\text{-}1)$$

Turboprop:
$$(Thr_{max\ climb})_{ISA} = \frac{C_{Tc,1}}{V_{TAS}} \times \left(1 - \frac{H_p}{C_{Tc,2}}\right) + C_{Tc,3} \quad (3.7\text{-}2)$$

Piston:
$$(Thr_{max\ climb})_{ISA} = C_{Tc,1} \times \left(1 - \frac{H_p}{C_{Tc,2}}\right) + \frac{C_{Tc,3}}{V_{TAS}} \quad (3.7\text{-}3)$$

For all engine types, the maximum climb thrust is corrected for temperature deviations from standard atmosphere, ΔT, in the following manner:

$$Thr_{max\ climb} = (Thr_{max\ climb})_{ISA} \times \left(1 - C_{Tc,5} \cdot \Delta T_{eff}\right) \quad (3.7\text{-}4)$$

Where:

$$\Delta T_{eff} = \Delta T - C_{Tc,4} \quad (3.7\text{-}5)$$

with the limits:

$$0.0 \leq \Delta T_{eff} \text{ x } C_{Tc,5} \leq 0.4 \quad (3.7\text{-}6)$$

and:

$$C_{Tc,5} \geq 0.0 \quad (3.7\text{-}7)$$

This maximum climb thrust is used for both take-off and climb phases.

3.7.2. Maximum Cruise Thrust

The normal cruise thrust is by definition set equal to drag (Thr = D). However, the maximum amount of thrust available in cruise situation is limited. The maximum cruise thrust is calculated as a ratio of the maximum climb thrust given by expression 3.7-4, that is:

$$(Thr_{cruise})_{MAX} = C_{Tcr} \times Thr_{max\ climb} \quad (3.7\text{-}8)$$

The coefficient C_{Tcr} is currently uniformly set for all aircraft (see Section 5.5).

3.7.3. Descent Thrust

Descent thrust is calculated as a ratio of the maximum climb thrust given by expression 3.7-4, with different correction factors used for high and low altitudes, and approach and landing configurations (see Section 3.5), that is:

if $H_p > H_{p,des}$:

$$Thr_{des,high} = C_{Tdes,high} \times Thr_{max\ climb} \quad (3.7\text{-}9)$$

if $H_p \leq H_{p,des}$:

Cruise configuration: $$Thr_{des,low} = C_{Tdes,low} \times Thr_{max\ climb} \quad (3.7\text{-}10)$$

Approach configuration: $$Thr_{des,app} = C_{Tdes,app} \times Thr_{max\ climb} \quad (3.7\text{-}11)$$

Landing configuration: $$Thr_{des,ld} = C_{Tdes,ld} \times Thr_{max\ climb} \quad (3.7\text{-}12)$$

Note that for those models where “non-clean” data (see Section 3.6.1) is available, $H_{p,des}$ cannot be below $H_{max,AP}$.

3.8. REDUCED CLIMB POWER

The reduced climb power has been introduced to allow the simulation of climbs using less than the maximum climb setting. In day-to-day operations, many aircraft use a reduced setting during climb in order to extend engine life and save cost. The correction factors that are used to calculate the reduction in power have been obtained in an empirical way and have been validated with the help of air traffic controllers.

In BADA, climbs that are performed using the full climb power will result in profiles that match the reference data that is found in the Flight Manual of the aircraft. Climbs with reduced power will give a realistic profile.

$$C_{pow,red} = 1 - C_{red} \times \frac{m_{max} - m_{act}}{m_{max} - m_{min}} \qquad (3.8\text{-}1)$$

The value of C_{red} is a function of the aircraft type and is given in Section 5.11.

Nevertheless:

If $H_p < (0.8 \cdot h_{max})$:

C_{red} = f (aircraft type) (see Section 5.11)

Else

$C_{red} = 0$ [dimensionless]

where h_{max} is given by expression 3.5-1.

The power reduction $C_{pow,red}$ is to be applied during the climb phase in expression 3.2-7, which becomes:

$$ROCD = \frac{dH_p}{dt} = \frac{T - \Delta T}{T} \frac{(Thr_{max\ climb} - D) \cdot V_{TAS} \cdot C_{pow,red}}{m \cdot g_0} \cdot f\{M\} \qquad \text{(in climb)} \qquad (3.8\text{-}2)$$

3.9. FUEL CONSUMPTION

3.9.1. Jet and Turboprop Engines

For the jet and turboprop engines, the thrust specific fuel consumption, η [kg/(min·kN)], is specified as a function of the true airspeed, V_{TAS} [kt]:

jet: $$\eta = C_{f1} \times \left(1 + \frac{V_{TAS}}{C_{f2}}\right) \quad (3.9\text{-}1)$$

turboprop: $$\eta = C_{f1} \times \left(1 - \frac{V_{TAS}}{C_{f2}}\right) \times \left(\frac{V_{TAS}}{1000}\right) \quad (3.9\text{-}2)$$

The nominal fuel flow, f_{nom} [kg/min], can then be calculated using the thrust, Thr:

jet/turboprop: $$f_{nom} = \eta \times Thr \quad (3.9\text{-}3)$$

These expressions are used in all flight phases except during idle descent and cruise, where the following expressions are to be used.

The minimum fuel flow, f_{min} [kg/min], corresponding to idle thrust descent conditions for both jet and turboprop engines, is specified as a function of the geopotential pressure altitude, H_p [ft], that is:

jet/turboprop: $$f_{min} = C_{f3} \left(1 - \frac{H_P}{C_{f4}}\right) \quad (3.9\text{-}4)$$

Note that for both jet and turboprop engines, the idle thrust part of the descent stops when the aircraft switches to approach and landing configuration (see Section 3.5), at which point thrust is generally increased. Hence, the calculation of fuel flow during approach and landing phases shall be based on the nominal fuel flow (expressions 3.7-11, 3.7-12 and 3.9-3), and limited to the minimum fuel flow (expression 3.9-4) if necessary:

jet/turboprop: $$f_{ap/ld} = MAX\ (f_{nom}, f_{min}) \quad (3.9\text{-}5)$$

The cruise fuel flow, f_{cr} [kg/min], is calculated using the thrust specific fuel consumption η, the thrust Thr, and a cruise fuel flow factor, C_{fcr}:

jet/turboprop: $$f_{cr} = \eta \times Thr \times C_{fcr} \quad (3.9\text{-}6)$$

For the moment the cruise fuel flow correction factor has been established for a number of aircraft types whenever the reference data for cruise fuel consumption is available. This factor has been set to 1 (one) for all the other aircraft models.

3.9.2. Piston Engines

For piston engines, the nominal fuel flow, f_{nom} [kg/min], is specified to be a constant, that is:

$$f_{nom} = C_{f1} \tag{3.9-7}$$

This expression is used in all flight phases except during descent and cruise, where the following expressions are to be used.

The minimum fuel flow, f_{min} [kg/min], corresponding to descent conditions for piston engines, is specified to be a constant:

$$f_{min} = C_{f3} \tag{3.9-8}$$

The cruise fuel flow, f_{cr} [kg/min], is calculated using a cruise fuel flow factor, C_{fcr}:

$$f_{cr} = C_{f1} \times C_{fcr} \tag{3.9-9}$$

For the moment the cruise fuel flow correction factor has been established for a number of aircraft types whenever the reference data for cruise fuel consumption is available. This factor has been set to 1 (one) for all the other aircraft models.

3.10. GROUND MOVEMENT

Four values are specified that can be of use when simulating ground movements. These parameters are:

- TOL: FAR Take-Off Length [m] with MTOW on a dry, hard, level runway under ISA conditions and no wind.
- LDL: FAR Landing Length [m] with MLW on a dry, hard, level runway under ISA conditions and no wind.
- span: aircraft wingspan [m]
- length: aircraft length [m]

Note that currently the value of the MLW is not provided in BADA. Apart from these model specific parameters, there are also a number of ground speeds defined as general parameters in Section 5.10.

3.11. SUMMARY OF OPERATIONS PERFORMANCE PARAMETERS

A summary of the parameters specified by the BADA operations performance model is supplied in Table 3-1 below. This table excludes those parameters that have been set to zero.

Detailed information on how these parameters have been obtained during the process of BADA aircraft model identification using the aircraft performance reference documents is provided in [RD6].

Important notice: Parameters listed in bold in the Table 3-1 below should not be modified by the user as such modifications may impact the validity of the data provided in [RD13].

Table 3-1: BADA Operations Performance Parameter Summary

Model Category	Symbols	Units	Description
Aircraft type (3 values)	**n_{eng}** **engine type** **wake category**	dimensionless string string	**number of engines** **either Jet, Turboprop or Piston** **either J, H, M or L**
Mass (4 values)	**m_{ref}** **m_{min}** **m_{max}** **m_{pyld}**	tonnes tonnes tonnes tonnes	**reference mass** **minimum mass** **maximum mass** **maximum payload mass**
Flight envelope (6 values)	**V_{MO}** **M_{MO}** **h_{MO}** **h_{max}** **G_w** **G_t**	knots (CAS) dimensionless feet feet feet/kg feet/K	**maximum operating speed** **maximum operating Mach number** **maximum operating altitude** **max. altitude at MTOW and ISA** **weight gradient on max. altitude** **temperature gradient on max. altitude**
Aerodynamics (16 values for jet aircraft, only 14 values for others)	**S** **$C_{D0,CR}$** **$C_{D2,CR}$** **$C_{D0,AP}$** **$C_{D2,AP}$** **$C_{D0,LD}$** **$C_{D2,LD}$** **$C_{D0,\Delta LDG}$** **$(V_{stall})_i$** **$C_{Lbo\ (M=0)}$**	m^2 dimensionless dimensionless dimensionless dimensionless dimensionless dimensionless dimensionless knots (CAS) dimensionless	**reference wing surface area** **parasitic drag coefficient (cruise)** **induced drag coefficient (cruise)** **parasitic drag coefficient (approach)** **induced drag coefficient (approach)** **parasitic drag coefficient (landing)** **induced drag coefficient (landing)** **parasite drag coef. (landing gear)** **stall speed [TO, IC, CR, AP, LD]** **Buffet onset lift coef. (jet only)**

Model Category	Symbols	Units	Description
	K	dimensionless	**Buffeting gradient (jet only)**
Engine thrust (12 values)	**$C_{Tc,1}$**	Newton (jet/piston) knot-Newton (turboprop)	**1st max. climb thrust coefficient**
	$C_{Tc,2}$	feet	**2nd max climb thrust coefficient**
	$C_{Tc,3}$	1/feet^2 (jet) Newton (turboprop) knot-Newton (piston)	**3rd max. climb thrust coefficient**
	$C_{Tc,4}$	K	**1st thrust temperature coefficient**
	$C_{Tc,5}$	1/K	**2nd thrust temperature coefficient**
	$C_{Tdes,low}$	dimensionless	**low altitude descent thrust coefficient**
	$C_{Tdes,high}$	dimensionless	**high altitude descent thrust coefficient**
	$H_{p,des}$	feet	**transition altitude for calculation of descent thrust**
	$C_{Tdes,app}$	dimensionless	**approach thrust coefficient**
	$C_{Tdes,ld}$	dimensionless	**landing thrust coefficient**
	$V_{des,ref}$	knots	reference descent speed (CAS)
	$M_{des,ref}$	dimensionless	reference descent Mach number
Fuel flow (5 values)	**C_{f1}**	kg/(min·kN) (jet) kg/(min·kN·knot) (turboprop) kg/min (piston)	**1st thrust specific fuel consumption coefficient**
	C_{f2}	knots	**2nd thrust specific fuel consumption coefficient**
	C_{f3}	kg/min	**1st descent fuel flow coefficient**
	C_{f4}	feet	**2nd descent fuel flow coefficient**
	C_{fcr}	dimensionless	**cruise fuel flow correction coefficient**
Ground movement (4 values)	TOL	m	take-off length
	LDL	m	landing length
	span	m	wingspan
	length	m	length

Note that the following coefficients can have negative values:
K, G_t, $C_{Tc,2}$, $C_{Tc,3}$, $C_{Tdes,low}$, $C_{Tdes,high}$, C_{f2}, C_{f4}.

4. AIRLINE PROCEDURE MODELS

This section defines the standard airline procedures, which are parameterised by the BADA airline procedure models. Definition of the standard airline procedures in BADA is driven by a requirement to provide means of simulating standard or nominal aircraft operations using different simulation and modelling tools for various ATM applications.

The BADA airline procedure model is provided for three separate flight phases: climb, cruise and descent. For each of these phases and each aircraft model, the BADA airline procedure model requires the following information to determine aircraft speed schedule:

1. BADA airline procedure default speeds provided in Airline Procedure File (APF):

   V_1 - standard CAS [knots] below 10,000 ft;

   V_2 - standard CAS [knots] between 10,000 ft and Mach transition altitude;

   M - standard Mach number above Mach transition altitude;

   where the Mach transition altitude is defined in Section 3.1 (g).
2. Stall speeds for take-off and landing configurations provided in Operations Performance File (OPF)
3. Coefficients provided in the Section 5.7 and 5.8

The process of definition of the BADA airline procedure default speeds and choice of aircraft configurations in function of flight phase is described in [RD6]. The airline procedure model below 10,000 ft with corresponding coefficients (mentioned under item 3 above) have been defined taking into account aircraft manufacturer's performance reference data and aircraft operational data available at EUROCONTROL.

The fact that the way aircraft is operated varies significantly in function of specific airspace procedures and operating policies of locally dominant airlines is widely recognised. It is for that reason that the resulting speed schedules of the BADA standard airline procedure model may differ from a geographical location or of an aerospace's specific aircraft operation.

To account for the local aircraft operation characteristics and improve conformance of the simulated aircraft behaviour with real operations, the user of BADA is given a possibility to modify the BADA default speeds (as provided in APF file). The change of speed related APF parameters should be done in accordance with the BADA modelling procedure described in the Chapter 2.2.3 of [RD6].

However, the stall speeds (as provided in OPF file) and coefficients detailed in Section 5.7 and 5.8 are not subject to modification. The BADA User should not modify them.

The altitude levels, used for determination of CAS speed schedules and provided in the following chapters, are expressed in terms of geopotential pressure altitude. However, different reference datums for altitude measurement[7] may be applied in function of the user application and its functional design choices.

The BADA Airline Procedure Model only identifies the possibility to introduce notion of different altitude altimetry for calculation of the CAS speed schedules in the user application. The implementation decision is left to the application owner.

[7] Such as use of standard operational pressure settings used in aviation: QNH for MSL pressure, QFE for pressure at the airport reference point or QNE corresponding to standard MSL1013 hPa.These can be selected through the altimeter's pressure setting knob in the aircraft.

4.1. CLIMB

The following parameters are defined for each aircraft type to characterise the climb phase:

$V_{cl,1}$ - standard climb CAS [knots] between 1,500/6,000 and 10,000 ft

$V_{cl,2}$ - standard climb CAS [knots] between 10,000 ft and Mach transition altitude

M_{cl} - standard climb Mach number above Mach transition altitude

- For jet aircraft the following CAS schedule is assumed, based on the parameters mentioned above and the take-off stall speed:

from 0 to 1,499 ft	$C_{Vmin} \cdot (V_{stall})_{TO} + Vd_{CL,1}$	(4.1-1)
from 1,500 to 2,999 ft	$C_{Vmin} \cdot (V_{stall})_{TO} + Vd_{CL,2}$	(4.1-2)
from 3,000 to 3,999 ft	$C_{Vmin} \cdot (V_{stall})_{TO} + Vd_{CL,3}$	(4.1-3)
from 4,000 to 4,999 ft	$C_{Vmin} \cdot (V_{stall})_{TO} + Vd_{CL,4}$	(4.1-4)
from 5,000 to 5,999 ft	$C_{Vmin} \cdot (V_{stall})_{TO} + Vd_{CL,5}$	(4.1-5)
from 6,000 to 9,999 ft	min ($V_{cl,1}$, 250 kt)	
from 10,000 ft to Mach transition altitude	$V_{cl,2}$	
above Mach transition altitude	M_{cl}	

- For turboprop and piston aircraft the following CAS schedule is assumed:

from 0 to 499 ft	$C_{Vmin} \cdot (V_{stall})_{TO} + Vd_{CL,6}$	(4.1-6)
from 500 to 999 ft	$C_{Vmin} \cdot (V_{stall})_{TO} + Vd_{CL,7}$	(4.1-7)
from 1,000 to 1,499 ft	$C_{Vmin} \cdot (V_{stall})_{TO} + Vd_{CL,8}$	(4.1-8)
from 1,500 to 9,999 ft	min ($V_{cl,1}$, 250 kt)	
from 10,000 ft to Mach transition altitude	$V_{cl,2}$	
above Mach transition altitude	M_{cl}	

Note 1: The take-off stall speed, $(V_{stall})_{TO}$, must be corrected for the difference in aircraft mass from the reference mass using formula 3.4-1. The values for $Vd_{CL,i}$ can be found in Section 5.

Note 2: The climb speed schedule shall determine an increasing speed from take-off to $V_{cl,1}$. To ensure that monotony, it is recommended to determine the speed schedule from the highest altitude to the lowest one, and to use at each step the speed of the higher altitude range as a ceiling value for the lower altitude range.

Note 3: Any speed from the schedule described above that would be lower (resp. higher) than the minimum (resp. maximum) speed determined for the same conditions using Section 3.5 (b) (resp. Section 3.5 (a)) shall be overriden by this minimum (resp. maximum) speed.

4.2. CRUISE

The following parameters are defined for each aircraft type to characterise the cruise phase:

$V_{cr,1}$ - standard cruise CAS [knots] between 3,000 and 10,000 ft
$V_{cr,2}$ - standard cruise CAS [knots] between 10,000 ft and Mach transition altitude
M_{cr} - standard cruise Mach number above Mach transition altitude

- For jet aircraft the following CAS schedule is assumed:

from 0 to 2,999 ft	min ($V_{cr,1}$, 170 kt)
from 3,000 to 5,999 ft	min ($V_{cr,1}$, 220 kt)
from 6,000 to 13,999 ft	min ($V_{cr,1}$, 250 kt)
from 14,000 ft to Mach transition altitude	$V_{cr,2}$
above Mach transition altitude	M_{cr}

- For turboprop and piston aircraft the following CAS schedule is assumed:

from 0 to 2,999 ft	min ($V_{cr,1}$, 150 kt)
from 3,000 to 5,999 ft	min ($V_{cr,1}$, 180 kt)
from 6,000 to 9,999 ft	min ($V_{cr,1}$, 250 kt)
from 10,000 ft to Mach transition altitude	$V_{cr,2}$
above Mach transition altitude	M_{cr}

Note: Any speed from the schedule described above that would be lower (resp. higher) than the minimum (resp. maximum) speed determined for the same conditions using Section 3.5 (b) (resp. Section 3.5 (a)) shall be overriden by this minimum (resp. maximum) speed.

4.3. DESCENT

The following parameters are defined for each aircraft type to characterise the descent phase:

$V_{des,1}$ - standard descent CAS [knots] between 3,000/6,000 and 10,000 ft
$V_{des,2}$ - standard descent CAS [knots] between 10,000 ft and Mach transition altitude
M_{des} - standard descent Mach number above Mach transition altitude

- For jet and turboprop aircraft the following CAS schedule is assumed, based on the above parameters and the landing stall speed:

from 0 to 999 ft	$C_{Vmin} \cdot (V_{stall})_{LD} + Vd_{DES,1}$	(4.3-1)
from 1,000 to 1,499 ft	$C_{Vmin} \cdot (V_{stall})_{LD} + Vd_{DES,2}$	(4.3-2)
from 1,500 to 1,999 ft	$C_{Vmin} \cdot (V_{stall})_{LD} + Vd_{DES,3}$	(4.3-3)
from 2,000 to 2,999 ft	$C_{Vmin} \cdot (V_{stall})_{LD} + Vd_{DES,4}$	(4.3-4)
from 3,000 to 5,999 ft	min ($V_{des,1}$, 220)	
from 6,000 to 9,999 ft	min ($V_{des,1}$, 250)	
from 10,000 ft to Mach transition altitude	$V_{des,2}$	
above Mach transition altitude	M_{des}	

- For piston aircraft the following CAS schedule is assumed:

from 0 to 499 ft	$C_{Vmin} \cdot (V_{stall})_{LD} + Vd_{DES,5}$	(4.3-5)
from 500 to 999 ft	$C_{Vmin} \cdot (V_{stall})_{LD} + Vd_{DES,6}$	(4.3-6)
from 1000 to 1,499 ft	$C_{Vmin} \cdot (V_{stall})_{LD} + Vd_{DES,7}$	(4.3-7)
from 1,500 to 9,999 ft	$V_{des,1}$	
from 10,000 ft to Mach transition altitude	$V_{des,2}$	
above Mach transition altitude	M_{des}	

Note 1: The landing stall speed, $(V_{stall})_{LD}$, must be corrected for the difference in aircraft mass from the reference mass using formula 3.4-1. The values for $Vd_{DES,i}$ can be found in Section 5.

Note 2: The descent speed schedule shall determine a decreasing speed from $V_{des,1}$ to landing. To ensure that monotony, it is recommended to evaluate the speed schedule from the highest altitude to the lowest one, and to use at each step the speed of the higher altitude range as a ceiling value for the lower altitude range.

Note 3: Any speed from the schedule described above that would be lower (resp. higher) than the minimum (resp. maximum) speed determined for the same conditions using Section 3.5 (b) (resp. Section 3.5 (a)) shall be overriden by this minimum (resp. maximum) speed.

5. GLOBAL AIRCRAFT PARAMETERS

5.1. INTRODUCTION

A number of parameters that have been described in Section 3 (Operations Performance Model) and Section 4 (Airline Procedure Model) have values that are independent of the aircraft type or model for which they are used. The values of these and other parameters which have general use, have been put in the Global Parameters File (BADA.GPF). This increases the flexibility and allows an easier evaluation of the values that are used.

The next section gives an overview of the parameters that are defined in the Global Parameters File. If relevant, it also indicates the formula in which the parameter should be used.

5.2. MAXIMUM ACCELERATION

Maximum acceleration parameters are used to limit the increment in TAS (longitudinal) or ROCD (normal). Two parameters are defined:

Name:	Description:	Value [ft/s^2]:
$a_{l,max\ (civ)}$	maximum longitudinal acceleration for civil flights	2.0
$a_{n,max\ (civ)}$	maximum normal acceleration for civil flights	5.0

The two acceleration limits are to be used in the following way:

- longitudinal acceleration: $\left| V_k - V_{k-1} \right| \leq a_{l,max\ (civ)}\ \Delta t$ (5.2-1)
- normal acceleration: $\left| \gamma_k - \gamma_{k-1} \right| \leq \frac{a_{n,max\ (civ)}\ \Delta t}{V}$ (5.2-2)

where,

$$\gamma = \sin^{-1}\left(\frac{\dot{h}}{V}\right) \quad (5.2\text{-}3)$$

and,

γ	is the climb/descent angle,
V	is the true airspeed [ft/s],
k, k-1	indicate values at update intervals k and k-1,
Δt	is the time interval between k and k-1 [s]

The values for the maximum longitudinal acceleration for military flights, $a_{l,max\ (mil)}$, and for the maximum normal acceleration for military flights, $a_{n,max\ (mil)}$, are currently undefined.

5.3. BANK ANGLES

Nominal and maximum bank angles are defined separately for military and civil flights. These bank angles can be used to calculate nominal and maximum rate of turns.

Name:	Description:	Value [deg]:
$\phi_{nom,civ\ (TO,LD)}$	Nominal bank angles for civil flight during TO and LD	15
$\phi_{nom,civ\ (OTHERS)}$	Nominal bank angles for civil flight during all other phases	35
$\phi_{nom,mil}$	Nominal bank angles for military flight (all phases)	50
$\phi_{max,civ\ (TO,LD)}$	Maximum bank angles for civil flight during TO and LD	25
$\phi_{max,civ\ (HOLD)}$	Maximum bank angles for civil flight during HOLD	35
$\phi_{max,civ\ (OTHERS)}$	Maximum bank angles for civil flight during all other phases	45
$\phi_{max,mil}$	Maximum bank angles for military flight (all phases)	70

The rate of turn, $\dot{\varphi}$, is calculated as a function of the bank angle:

$$\dot{\varphi} = \frac{g_0}{V_{TAS}} \times \tan(\phi) \qquad (5.3\text{-}1)$$

5.4. EXPEDITED DESCENT

The expedited descent factor is to be used as a drag multiplication factor during expedited descents in order to simulate use of spoilers:

Name:	Description:	Value [-]:
$C_{des,exp}$	Expedited descent factor	1.6

The drag during an expedited descent is calculated using the nominal drag (see Section 3.6.1):

$$D_{des,exp} = C_{des,exp} \cdot D_{nom} \qquad (5.4\text{-}1)$$

5.5. THRUST FACTORS

Maximum take-off and maximum cruise thrust factors have been specified. The $C_{Th,TO}$ factor is no longer used since BADA 3.0. The C_{Tcr} factor is to be used in expression 3.7-8.

Name:	Description:	Value [-]:
$C_{Th,TO}$	Take-off thrust coefficient	1.2
C_{Tcr}	Maximum cruise thrust coefficient	0.95

5.6. CONFIGURATION ALTITUDE THRESHOLD

For 4 configurations, altitude thresholds have been specified in BADA: take-off (TO), initial climb (IC), approach (AP) and landing (LD). Note that the selection of the take-off and initial climb configurations is defined only with the altitude. The selection of the approach and landing configurations is done through the use of air speed and altitude (see Section 3.5), while the altitudes at which the configuration change takes place should not be higher than the ones given below. The altitude values are expressed in terms of geopotential pressure altitude.

Name:	Description:	Value [ft]:
$H_{max,TO}$	Maximum altitude threshold for take-off	400
$H_{max,IC}$	Maximum altitude threshold for initial climb	2,000
$H_{max,AP}$	Maximum altitude threshold for approach	8,000
$H_{max,LD}$	Maximum altitude threshold for landing	3,000

5.7. MINIMUM SPEED COEFFICIENTS

Two minimum speed coefficients are specified, which are to be used in expressions 3.5-2 and 3.5-3 and (for C_{Vmin} only) in Section 4.1 and 4.3:

Name:	Description:	Value [-]:
$C_{Vmin,TO}$	Minimum speed coefficient for take-off	1.2
C_{Vmin}	Minimum speed coefficient (all other phases)	1.3

5.8. SPEED SCHEDULES

The speed schedules applicable below FL100 for climb and descent are based on a factored stall speed plus increment valid for a specified geopotential pressure altitude range.

Name:	Description:	Value [KCAS]:
$Vd_{CL,1}$	Climb speed increment below 1500 ft (jet)	5
$Vd_{CL,2}$	Climb speed increment below 3000 ft (jet)	10
$Vd_{CL,3}$	Climb speed increment below 4000 ft (jet)	30
$Vd_{CL,4}$	Climb speed increment below 5000 ft (jet)	60
$Vd_{CL,5}$	Climb speed increment below 6000 ft (jet)	80
$Vd_{CL,6}$	Climb speed increment below 500 ft (turbo/piston)	20
$Vd_{CL,7}$	Climb speed increment below 1000 ft (turbo/piston)	30
$Vd_{CL,8}$	Climb speed increment below 1500 ft (turbo/piston)	35
$Vd_{DES,1}$	Descent speed increment below 1000 ft (jet/turboprop)	5
$Vd_{DES,2}$	Descent speed increment below 1500 ft (jet/turboprop)	10
$Vd_{DES,3}$	Descent speed increment below 2000 ft (jet/turboprop)	20
$Vd_{DES,4}$	Descent speed increment below 3000 ft (jet/turboprop)	50
$Vd_{DES,5}$	Descent speed increment below 500 ft (piston)	5

$Vd_{DES,6}$	Descent speed increment below 1000 ft (piston)	10
$Vd_{DES,7}$	Descent speed increment below 1500 ft (piston)	20

These values are to be used in the expressions in Section 4.1 and 4.3.

5.9. HOLDING SPEEDS

The holding speeds that are to be used to calculate holding areas are defined according to the ICAO standards:

Name:	Description:	Value [KCAS]:
$V_{hold,1}$	Holding speed below FL140	230
$V_{hold,2}$	Holding speed between FL140 and FL200	240
$V_{hold,3}$	Holding speed between FL200 and FL340	265
$V_{hold,4}$	Holding speed above FL340 [Mach]	0.83

Note that the holding speeds that are used by individual aircraft may vary between types.

5.10. GROUND SPEEDS

A number of ground speeds are defined for the simulation of ground movement. For the moment, no distinction between aircraft type or engine type is made. The following speeds have been defined:

Name:	Description:	Value [KCAS]:
$V_{backtrack}$	Runway backtrack speed	35
V_{taxi}	Taxi speed	15
V_{apron}	Apron speed	10
V_{gate}	Gate speed	5

The runway backtrack speed is the speed the aircraft will maintain when it backtracks across the runway. The taxi speed is used anywhere between the runway and the apron area. The apron speed is used in the apron area while the gate speed is used for all manoeuvring between the gate position and the apron.

5.11. REDUCED POWER COEFFICIENT

The reduced power coefficients are defined for the three different engine types. It is stressed that the values given below were found in an empirical way and have been validated with the help of air traffic controllers:

Name:	Description:	Value [-]:
$C_{red,turbo}$	Maximum reduction in power for turboprops	0.25
$C_{red,piston}$	Maximum reduction in power for pistons	0.0
$C_{red,jet}$	Maximum reduction in power for jets	0.15

The coefficients should be used in Formula 3.8-1.

6. FILE STRUCTURE

6.1. FILE TYPES

All data provided by BADA Revision 3.11 are organised into six types of files:

- three Synonym Files,
- a set of Operations Performance Files,
- a set of Airline Procedure Files,
- a set of Performance Table Files,
- a set of Performance Table Data,
- a Global Parameter File.

- Three Synonym Files have the names:

  SYNONYM.LST
  SYNONYM.NEW
  SYNONYM_ALL.LST

The files provide a list of all the aircraft types which are supported by BADA and indicate whether the aircraft type is supported directly (through provision of parameters in other files) or supported by equivalence (through indicating an equivalent aircraft type that is supported directly). In addition to that, SYNONYM_ALL.LST file provides the information on history and evolution of the ICAO aircraft designators over the years. The format of the files is described in Section 6.3.

- There is one Operations Performance File (OPF) provided for each aircraft type which is directly supported. This file specifies parameter values for the mass, flight envelope, drag, engine thrust and fuel consumption that are described in Section 3. Details on the format of the OPF file are given in Section 6.4.

- There is one Airline Procedures File (APF) for each directly supported aircraft type. This file specifies the nominal manoeuvre speeds that are described in Section 4. Details on the format of the APF file are given in Section 6.5.

- There is one Performance Table File (PTF) for each directly supported aircraft type. This file contains a summary table of speeds, climb/descent rates and fuel consumption at various flight levels. Details on the format of the PTF file are given in Section 6.6.

- There is one Performance Table Data (PTD) file for each directly supported aircraft type. This file contains a detailed table of computed performance values at various flight levels. Details on the format of the PTD file are given in Section 6.7.

- Finally there is one Global Parameter File which is named BADA.GPF. This file contains parameters that are described in Section 5 and are valid for all aircraft or a group of aircraft (for instance all civil flights or all jet aircraft). Details on the format of the GPF file are given in Section 6.8.

The names of the OPF, APF, PTF and PTD files are based on the ICAO designation code for the aircraft type. With only the exception of the generic military fighter aircraft types (FGTH, FGTL, FGTN), this code is the same as the International Civil Aviation Organisation (ICAO) designator code for the aircraft type [RD2]. That is:

Operations Performance File name:	<ICAO_code>__.OPF
Airline Procedures File name:	<ICAO_code>__.APF
Performance Table File name:	<ICAO_code>__.PTF
Performance Table Data name:	<ICAO_code>__.PTD

Note that there are at least two underscore characters between the ICAO code and the file extension such that the length of the file name without the extension is six characters. Most ICAO codes are four characters in length and thus have two underscore characters. Some ICAO codes, however, can be shorter (e.g. F50) and thus require more underscore characters. For example, an Airbus 310 which has the ICAO code of A310 is represented in BADA by the following files:

Operations Performance File:	A310__.OPF
Airline Procedures File:	A310__.APF
Performance Table File:	A310__.PTF
Performance Table Data:	A310__.PTD

The Fokker F50, which has the ICAO code of F50, is represented in BADA by the following files:

Operations Performance File:	F50____.OPF
Airline Procedures File:	F50____.APF
Performance Table File:	F50____.PTF
Performance Table Data:	F50____.PTD

All files belonging to BADA Revision 3.11, that is the Synonym Files, the GPF file and all APF, OPF, PTF and PTD files, are controlled within a configuration management system. This system is described in Section 6.2.

6.2. FILE CONFIGURATION MANAGEMENT

Starting with the BADA 3.4 release, the BADA Synonym Files, GPF and all APF, OPF, PTF and PTD files are placed and managed under the Change Management Synergy (CM Synergy) tool at EUROCONTROL.

This section briefly describes some of the CM Synergy features that will be used for the management of the BADA files.

CM Synergy provides a complete change management environment in which development and management of the files can be done easily, quickly, and securely. It maintains control of file versions and allows management of project releases with some of the benefits listed below:

- workflow management, which enables easy identification of the files modified to implement the change, and review of the reason for a change,
- project reproducibility by accurately creating baseline configurations,

- role-based security,

- Distributed Change Management (DCM) which allows files sharing among any number of CM Synergy databases. With DCM transfer of an entire database or a subset of a database can be done, either automatically or manually.

The CM Synergy automated migration facilities feature complete version history migration from RCS system archives. This has enabled to bring successfully all the BADA files with their history under the CM Synergy control. A CM Synergy database is created for BADA project. Such a database represents a data repository that stores all controlled data, including data files, their properties and relationships to one another.

The following BADA files are placed in the CM Synergy database:

- the Synonym Files
- the GPF file
- all APF, OPF, PTF and PTD files

Within the CM Synergy, different methodologies in the way the files are managed are used. For BADA database, the task-based methodology is chosen which enables the tracking of the changes by using tasks, rather than individual files, as the basic unit of work.

The specific procedures used for configuration management are specified in the BADA Configuration Management Manual [RD5].

6.2.1. File Identification

Any file managed in a CM Synergy database is uniquely identified by the following attributes: *name, version, type, and instance.* By default, the four-part name (also called full name) is written like this: *name-version:type:instance*.

A file name can be up to 151 characters long, and the version can be any 32-character combination. The type can be any of the default types (e.g., *csrc, ascii*, etc.), or any BADA type that is created (APF, OPF, PTF, PTD, GPF).

The name, version, and type are designated by the user, but the instance is calculated by CM Synergy.

The version of a file corresponds to the evolution of the file in time. By default, CM Synergy creates version numbers, starting with 1, for each file that is created in the CM Synergy database. Each time the object is modified, CM Synergy increments the version.

The instance is used to distinguish between multiple objects with the same name and type, but that are not versions of each other.

It is important to notice that, following the CM Synergy approach of the file identification, no information on the file version is provided in the BADA file itself.

A new layout of the header of BADA files has been developed and it will be described in more details in the following sections.

6.2.2. History

The history of a file shows all the existing versions and the relationships between the versions. By history, CM Synergy means all of the file versions created before the current file version (called predecessors) and all of the file versions created after the current file version (called successors). This functionality allows for the tracking of all modifications to a file.

6.2.3. Release

The release is a label that indicates the version of the project, in this case the release of BADA files. BADA releases are usually identified by a two digit number, e.g. 3.3 or 3.4. However, the name of release in CM Synergy can be made out of any combination of alphabetic and numerical characters.

Like in the case of the file version, no information on the current BADA release is given in the BADA files.

6.2.4. Release Summary file

The ReleaseSummary file provides a list of all files delivered as part of the BADA release. It lists, for each BADA file, the file name and BADA release identification, which is the BADA release in which the file was last modified.

6.3. SYNONYM FILE FORMAT

6.3.1. SYNONYM.LST File

The SYNONYM.LST file is an ASCII file which lists all aircraft types which are supported by the BADA revision. An example of the SYNONYM.LST file is given below (partial listing).

```
CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC SYNONYM.LST CCCCCCCCCCCCC/
CC                                                                    /
CC                BADA SYNONYM FILE                                   /
CC                                                                    /
CC                                                                    /
CC      File_name: SYNONYM.LST                                        /
CC                                                                    /
CC      Creation_date: Mar 26 2002                                    /
CC                                                                    /
CC      Modification_date: Mar 26 2002                                /
CC                                                                    /
CC                                                                    /
CC====== Aircraft List ================================================/
CC                                                                    /
CC   A/C    NAME OR MODEL        FILE               SYNONYMS         /
CC   CODE                                                            /
CC                                                                    /
 - A306__  AIRBUS A300B4-600    A306__             A306
 - A30B__  AIRBUS A300B4-200    A30B__             A30B   IL76
 - A310__  AIRBUS A310          A310__             A310
 - A319__  AIRBUS A319          A319__             A319
 - A320__  AIRBUS A320          A320__             A320   C17
 - A321__  AIRBUS A321          A321__             A321
 - A333__  AIRBUS A330-300      A333__             A333   A332
 - A343__  AIRBUS A340-300      A343__             A343   A342   A345
                                                          A346
 - AT43__  ATR ATR 42-300       AT43__             AT43   CN35   CVLT
                                                   AT44
 - AT45__  ATR ATR 42-500       AT45__             AT45
 - AT72__  ATR ATR 72           AT72__             AT72   A748
 - ATP___  ADVANCED TURBOPROP   ATP___             ATP    G222
 - B461__  BAE 146-100/RJ       B461__             B461   B462   B463
                                                   YK42
 - B703__  BOEING 707-300       B703__             B703   B720   K35R
                                                   E3TF   E3CF   C135
                                                   VC10   IL62
 - B722__  BOEING 727-100       B722__             B722   B721   BER4
 - B732__  BOEING 737-228       B732__             B732   B731   A124
 - B733__  BOEING 737-300       B733__             B733
 - B734__  BOEING 737-400       B734__             B734
 - B735__  BOEING 737-500       B735__             B735   B736
 - B737__  BOEING 737-700       B737__             B737
```

There are three types of lines in the SYNONYM.LST file with the line type identified by the first two characters in the line. These line types with their associated leading characters are listed below.

CC	comment line
CD	data line
-	synonym line

The data is organised into two blocks separated by a comment line consisting of the block name and equal signs "=":

- file identification block
- aircraft list block

Each of these blocks is described in the subsections below.

6.3.1.1. File Identification Block

The file identification block provides information on the file name, creation and modification date. The block consists of 12 comment lines.

```
CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC SYNONYM.LST CCCCCCCCCCCCCC/
CC                                                                     /
CC                BADA SYNONYM FILE                                    /
CC                                                                     /
CC                                                                     /
CC      File_name: SYNONYM.LST                                         /
CC                                                                     /
CC      Creation_date: Mar 26 2002                                     /
CC                                                                     /
CC      Modification_date: Mar 26 2002                                 /
```

The comment lines specify the file name along with the creation and the modification date. The creation date indicates the date when the file was created for the first time. The modification date indicates when the contents of the file were last modified.

6.3.1.2. Aircraft Listing Block

The aircraft listing block consists of 5 comment lines with additional synonym lines for each aircraft supported by the BADA Revision. A partial listing of this block is shown below.

```
CC====== Aircraft List ============================================/
CC                                                                 /
CC   A/C    NAME OR MODEL        FILE               SYNONYMS       /
CC   CODE                                                          /
CC                                                                 /
 - A306__  AIRBUS A300B4-600    A306__             A306
 - A30B__  AIRBUS A300B4-200    A30B__             A30B   IL76
 - A310__  AIRBUS A310          A310__             A310
 - A319__  AIRBUS A319          A319__             A319
 - A320__  AIRBUS A320          A320__             A320   C17
 - A321__  AIRBUS A321          A321__             A321
 - A333__  AIRBUS A330-300      A333__             A333   A332
 - A343__  AIRBUS A340-300      A343__             A343   A342
 - AT43__  ATR ATR 42-300       AT43__             AT43   CN35   CVLT
                                                   AT44
 - AT45__  ATR ATR 42-500       AT45__             AT45
 - AT72__  ATR ATR 72           AT72__             AT72   A748
 - ATP___  ADVANCED TURBOPROP   ATP___             ATP    G222
 - B461__  BAE 146-100/RJ       B461__             B461   B462   B463
                                                   YK42
 - B703__  BOEING 707-300       B703__             B703   B720   K35R
                                                   E3TF   E3CF   C135
                                                   VC10   IL62
 - B722__  BOEING 727-100       B722__             B722   B721   BER4
 - B732__  BOEING 737-228       B732__             B732   B731   A124
 - B733__  BOEING 737-300       B733__             B733
 - B734__  BOEING 737-400       B734__             B734
 - B735__  BOEING 737-500       B735__             B735   B736
 - B737__  BOEING 737-700       B737__             B737
```

There is one synonym line for each of the directly supported aircraft within the BADA release. Each such line consists of 4 fields as described below:

(a) Aircraft Code Field

This field identifies the aircraft type. It consists of a three or four-character ICAO code followed by two or more underscore characters.

(b) Name or Model Field

This field identifies the manufacturer and model of the aircraft.

(c) File Name Field

This field identifies the file name for the APF, OPF, PTF or PTD files associated with the aircraft (minus the file extension). For each aircraft this is the same as the A/C code.

(d) Equivalence Field

This field lists any equivalences associated with the aircraft. By default, each aircraft has at least one equivalence to itself.

Note that in some cases the name or model or equivalence fields may be continued onto the next line as it is the case with the B703 model.

6.3.2. SYNONYM.NEW File

The SYNONYM.NEW file is an ASCII file, which lists all aircraft types, which are supported by the BADA revision. Its format differs from the SYNONYM.LST file in that all supported aircraft are listed alphabetically in the file whether they are supported directly or by equivalence. An example of the SYNONYM.NEW file is given below (partial listing).

```
CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC SYNONYM.NEW CCCCCCCCCCCCCC/
CC                                                                    /
CC                 BADA SYNONYM FILE                                  /
CC                                                                    /
CC                                                                    /
CC      File_name: SYNONYM.NEW                                        /
CC                                                                    /
CC      Creation_date: Mar 26 2002                                    /
CC                                                                    /
CC      Modification_date: Mar 26 2002                                /
CC                                                                    /
CC                                                                    /
CC====== Aircraft List ================================================/
CC                                                                    /
CC   A/C    MANUFACTURER        NAME OR MODEL            FILE    OLD  /
CC   CODE                                                        CODE /
CC                                                                    /
CD * A10    FAIRCHILD           THUNDERBOLT II           FGTN__  A10A /
CD * A124   ANTONOV             ANTONOV AN-124           B732__  AN4R /
CD - A306   AIRBUS              A300B4-600               A306__  A306 /
CD - A30B   AIRBUS              A300B4-200               A30B__  A300 /
CD - A310   AIRBUS              A310                     A310__  A310 /
CD - A319   AIRBUS              A319                     A319__  A319 /
CD - A320   AIRBUS              A320                     A320__  EA32 /
CD - A321   AIRBUS              A321                     A321__  A321 /
CD * A332   AIRBUS              A330-200                 A333__  A332 /
CD - A333   AIRBUS              A330-300                 A333__  A330 /
CD * A342   AIRBUS              A340-200                 A343__  A342 /
CD - A343   AIRBUS              A340-300                 A343__  A340 /
CD * A345   AIRBUS              A340-500                 A343__  A345 /
CD * A346   AIRBUS              A340-600                 A343__  A346 /
CD * A4     MCDONNELL-DOUGLAS   SKYHAWK                  FGTN__  A4   /
CD * A6     GRUMMAN             INTRUDER                 FGTN__  EA6B /
CD * A748   BAE                 BAE 748                  AT72__  HN74 /
CD * AC80   ROCKWELL            TURBO COMMANDER          BE20__  AC6T /
CD * AC90   ROCKWELL            TURBO COMMANDER          BE20__  AC90 /
CD * AC95   ROCKWELL            TURBO COMMANDER          BE20__  AC95 /
CD * AJET   DASSAULT            ALPHA JET                FGTN__  AJET /
CD * AMX    EMBRAER             AMX                      FGTN__  AMX  /
CD * AN12   ANTONOV             AN-12                    C130__  AN12 /
CD * AN24   ANTONOV             AN-124                   F27___  AN24 /
CD * AN26   ANTONOV             AN-26                    F27___  AN26 /
```

There are three types of lines in the SYNONYM.NEW file with the line type identified by the first two characters in the line. These line types with their associated two leading characters are listed below.

CC	comment line
CD	data line
FI	end-of-file line

The data is organised into two blocks separated by a comment line consisting of the block name and equal signs "=":

- file identification block
- aircraft list block

Each of these blocks is described in the subsections below.

6.3.2.1. File Identification Block

The file identification block provides information on the file name, creation and modification date. The block consists of 12 comment lines as shown below.

```
CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC SYNONYM.NEW CCCCCCCCCCCCCCC/
CC                                                                    /
CC                BADA SYNONYM FILE                                   /
CC                                                                    /
CC                                                                    /
CC      File_name: SYNONYM.NEW                                        /
CC                                                                    /
CC      Creation_date: Mar 26 2002                                    /
CC                                                                    /
CC      Modification_date: Mar 26 2002                                /
CC                                                                    /
```

The comment lines specify the file name along with the creation and last modification date. The creation date indicates the date when the file was created for the first time. The modification date indicates when the contents of the file were last modified.

6.3.2.2. Aircraft Listing Block

The aircraft listing block consists of 5 comment lines and at least one data line for each aircraft supported by the BADA Revision. Some aircraft have more than one data line, see under (f). A partial listing of this block is shown below.

```
CD * A10    FAIRCHILD          THUNDERBOLT II          FGTN__  A10A /
CD * A124   ANTONOV            ANTONOV AN-124          B732__  AN4R /
CD - A306   AIRBUS             A300B4-600              A306__  A306 /
CD - A30B   AIRBUS             A300B4-200              A30B__  A300 /
```

Each data line consists of 6 fields as described below:

(a) Support Type Field

This field is one character in length being one of the following two values:

"-" to indicate an aircraft type directly supported, and,

"*" to indicate an aircraft type supported by equivalence with another directly supported aircraft

(b) Aircraft Code Field

This field identifies the aircraft type. It consists of a three or four-character ICAO code.

(c) Manufacturer Field

This field identifies the manufacturer of the aircraft. Examples are Boeing, Airbus or Fokker.

(d) Name or Model Field

This field identifies the name or model for the aircraft type. Examples are the 747-400 series or Learjet 35.

(e) File Field

This field indicates the name of the OPF, APF, PTF or PTD file, which contains the parameters for the aircraft type (minus the file extension).

For an aircraft type which is directly supported this file name will be the same as the ICAO code with an additional two or more underscore characters to form a string of six characters in length. For example, the file name corresponding to the A333 will be A333__. This indicates an OPF file A333__.OPF, an APF file A333__.APF, a PTF file A333__.PTF and a PTD file A333__.PTD. For the Fokker F-27 with an ICAO code of F27, the file names include three underscore characters, that is, F27___.OPF, F27___.APF, F27___.PTF and F27___.PTD.

For an aircraft type which is supported through equivalence the file name will indicate the file for the equivalent aircraft type which should be used. As an example, the Antonov 12 (AN12) is equivalent to the Lockheed C-130 Hercules (C130). Thus the files C130__.OPF, C130__.APF, C130__.PTF and C130__.PTD should be used.

(f) Old Code field

The old code field gives the name of the aircraft that refers to the formerly known aircraft designator as published in one of the previous editions of the ICAO document 8643 [RD10]. This allows the BADA Revision 3.11 user to continue to use the old ICAO standard and to establish a link between the old and the new aircraft designators.

The above fields are specified in the following fixed format (Fortran notation):

'CD', 1X, A1, 1X, A4, 3X, A18, 1X, A25, 1X, A6, 2X, A4

6.3.3. SYNONYM_ALL.LST File

The SYNONYM_ALL.LST file is an ASCII file, which lists all aircraft types, which are supported by the BADA revision. Like in the SYNONYM.NEW file, all supported aircraft are listed alphabetically in the file whether they are supported directly or by equivalence. An example of the SYNONYM_ALL.LST file is given below (partial listing).

```
CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC SYNONYM_ALL.LST CCCCCCCCCCCCCCC/
CC                                                                     /
CC                BADA SYNONYM_ALL FILE                                /
CC                                                                     /
CC                                                                     /
CC      File_name: SYNONYM_ALL.LST                                     /
CC                                                                     /
CC      Creation_date: May 22 2003                                     /
CC                                                                     /
CC      Modification_date: May 22 2003                                 /
CC                                                                     /
CC                                                                     /
CC                                                                     /
CC=====================================================================/
CC                                                                     /
CC                                                                     /
CC                                                        ICAO    ICAO    ICAO    ICAO    ICAO    ICAO    ICAO
CC   A/C    NAME OR MODEL    MANUFACTURER  FILE           CODE    CODE    CODE    CODE    CODE    CODE    CODE
CC                                        BADA 3.5        V24     V25     V26     V27     V28     V29     V30
CC
CD * A10    THUNDERBOLT II    FAIRCHILD    FGTN__       A10A     A10     A10     A10     A10     A10     A10
CD * A124   ANTONOV AN-124     ANTONOV      B732__       AN4R     A124    A124    A124    A124    A124    A124
CD - A306   A300B4-600        AIRBUS       A306__       A306     A306    A306    A306    A306    A306    A306
CD - A30B   A300B4-200        AIRBUS       A30B__       EA30     A300    A30B    A30B    A30B    A30B    A30B
CD - A310   A310              AIRBUS       A310__       EA31     A310    A310    A310    A310    A310    A310
CD * A318   A318              AIRBUS       A319__       A318     A318    A318    A318    A318    A318    A318
CD - A319   A319              AIRBUS       A319__       A319     A319    A319    A319    A319    A319    A319
CD - A320   A320              AIRBUS       A320__       EA32     A320    A320    A320    A320    A320    A320
CD - A321   A321              AIRBUS       A321__       A321     A321    A321    A321    A321    A321    A321
CD - A332   A330-200          AIRBUS       A332__       A332     A332    A332    A332    A332    A332    A332
CD - A333   A330-300          AIRBUS       A333__       EA33     EA33    EA33    A330    A333    A333    A333
CD * A342   A340-200          AIRBUS       A343__       A342     A342    A342    A342    A342    A342    A342
CD - A343   A340-300          AIRBUS       A343__       EA34     EA34    EA34    A340    A343    A343    A343
CD * A345   A340-500          AIRBUS       A343__       A345     A345    A345    A345    A345    A345    A345
CD * A346   A340-600          AIRBUS       A343__       A346     A346    A346    A346    A346    A346    A346
```

There are three types of lines in the SYNONYM_ALL.LST file with the line type identified by the first two characters in the line. These line types with their associated two leading characters are listed below.

CC	comment line
CD	data line
FI	end-of-file line

The data is organised into two blocks separated by a comment line consisting of equal signs "=":

- file identification block
- aircraft list block

Each of these blocks is described in the subsections below.

6.3.3.1. File Identification Block

The file identification block provides information on the file name, creation and modification date. The block consists of 13 comment lines as shown below.

```
CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC SYNONYM_ALL.LST CCCCCCCCCCCCCCC/
CC                                                                     /
CC                BADA SYNONYM_ALL FILE                                /
CC                                                                     /
CC                                                                     /
CC      File_name: SYNONYM.NEW                                         /
CC                                                                     /
CC      Creation_date: May 22 2003                                     /
CC                                                                     /
CC      Modification_date: May 22 2003                                 /
CC                                                                     /
```

The comment lines specify the file name along with the creation and last modification date. The creation date indicates the date when the file was created for the first time. The modification date indicates when the contents of the file were last modified.

6.3.3.2. Aircraft Listing Block

The aircraft listing block consists of 7 comment lines and at least one data line for each aircraft supported by the BADA Revision. Some aircraft have more than one data line, see under (f). A partial listing of this block is shown below.

```
* A10    THUNDERBOLT II    FAIRCHILD    FGTN__    A10A    A10    A10    A10    A10    A10    A10
* A124   ANTONOV AN-124    ANTONOV      B732__    AN4R    A124   A124   A124   A124   A124   A124
- A306   A300B4-600        AIRBUS       A306__    A306    A306   A306   A306   A306   A306   A306
- A30B   A300B4-200        AIRBUS       A30B__    EA30    A300   A30B   A30B   A30B   A30B   A30B
```

Each data line consists of 5 fields describing the aircraft type and number of additional fields providing the history of ICAO aircraft type designators. Detailed description is given below:

(a) Support Type Field

This field is one character in length being one of the following two values:

"-" to indicate an aircraft type directly supported, and,

"*" to indicate an aircraft type supported by equivalence with another directly supported aircraft

(b) Aircraft Code Field

This field identifies the aircraft type. It consists of a three or four-character ICAO code.

(c) Manufacturer Field

This field identifies the manufacturer of the aircraft. Examples are Boeing, Airbus or Fokker.

(d) Name or Model Field

This field identifies the name or model for the aircraft type. Examples are the 747-400 series or Learjet 35.

(e) File Field

This field indicates the name of the OPF, APF, PTF or PTD file, which contains the parameters for the aircraft type (minus the file extension).

For an aircraft type which is directly supported this file name will be the same as the ICAO code with an additional two or more underscore characters to form a string of six characters in length. For example, the file name corresponding to the A333 will be A333__. This indicates an OPF file A333__.OPF, an APF file A333__.APF, a PTF file A333__.PTF and a PTD file A333__.PTD. For the Fokker F-27 with an ICAO code of F27, the file names include three underscore characters, that is, F27___.OPF, F27___.APF, F27___.PTF and F27___.PTD.

For an aircraft type which is supported through equivalence the file name will indicate the file for the equivalent aircraft type which should be used. As an example, the Antonov 12 (AN12) is equivalent to the Lockheed C-130 Hercules (C130). Thus the files C130__.OPF, C130__.APF, C130__.PTF and C130__.PTD should be used.

(f) Old Code fields

The old code fields give the name of the aircraft that refers to the formerly known aircraft designator as published in one of the previous editions (versions, i.e. V24 to V37) of the ICAO document 8643 [RD10]. This allows the BADA Revision 3.11 user to continue to use the old ICAO standard and to establish a link between the old and the new aircraft designators, as well as the corresponding aircraft file name from the most recent BADA release. If the specific aircraft model version did not have an assigned designator in the past editions of the ICAO document or the information was not available to the BADA team, then the most recent designator is used throughout all the versions.

6.4. OPF FILE FORMAT

The Operations Performance File (OPF) is an ASCII file, which for a particular aircraft type specifies the operations performance parameters described in Section 3. An example of an OPF file for the A306 (Airbus 300B4-600) aircraft is shown below.

```
CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC A306__.OPF CCCCCCCCCCCCCC/
CC                                                                          /
CC                AIRCRAFT PERFORMANCE OPERATIONAL FILE                     /
CC                                                                          /
CC                                                                          /
CC      File_name: A306__.OPF                                               /
CC                                                                          /
CC      Creation_date: Mar 26 2002                                          /
CC                                                                          /
CC      Modification_date: Mar 26 2002                                      /
CC                                                                          /
CC                                                                          /
CC====== Actype ============================================================/
CD   A306__         2 engines    Jet                        H               /
CC   Airbus A300-B4-622 with PW4158  engines              wake              /
CC                                                                          /
CC====== Mass (t) ==========================================================/
CC    reference      minimum      maximum     max payload  mass grad  /
CD     .14000E+03   .87000E+02   .17170E+03   .39000E+02   .14100E+00 /
CC====== Flight envelope ===================================================/
CC     VMO(KCAS)       MMO        Max.Alt       Hmax       temp grad  /
CD     .33500E+03   .82000E+00   .41000E+05   .31600E+05  -.67000E+02 /
CC====== Aerodynamics ======================================================/
CC Wing Area and Buffet coefficients (SIM)                                  /
CCndrst Surf(m2)     Clbo(M=0)       k           CM16                       /
CD 5   .26000E+03   .15300E+01   .10290E+01   .00000E+00                   /
CC   Configuration characteristics                                          /
CC n Phase  Name    Vstall(KCAS)    CD0          CD2        unused    /
CD 1 CR   Clean     .15100E+03   .19000E-01   .53000E-01   .00000E+00 /
CD 2 IC   S15F00    .11700E+03   .33057E-01   .45362E-01   .00000E+00 /
CD 3 TO   S15F00    .11700E+03   .33057E-01   .45362E-01   .00000E+00 /
CD 4 AP   S15F15    .10900E+03   .38031E-01   .44932E-01   .00000E+00 /
CD 5 LD   S30F40    .97000E+02   .78935E-01   .44822E-01   .00000E+00 /
CC   Spoiler                                                               /
CD 1      RET                                                              /
CD 2      EXT                                .00000E+00   .00000E+00 /
CC   Gear                                                                  /
CD 1      UP                                                               /
CD 2      DOWN                  .22500E-01   .00000E+00   .00000E+00 /
CC   Brakes                                                                /
CD 1      OFF                                                              /
CD 2      ON                                 .00000E+00   .00000E+00 /
CC====== Engine Thrust =====================================================/
CC         Max climb thrust coefficients (SIM)                              /
CD     .30400E+06   .44800E+05   .11600E-09   .67500E+01   .42600E-02 /
CC      Desc(low)    Desc(high)   Desc level   Desc(app)    Desc(ld)  /
CD     .73000E-02   .20600E-01   .80000E+04   .12000E+00   .36000E+00 /
CC      Desc CAS     Desc Mach    unused       unused       unused    /
CD     .28000E+03   .79000E+00   .00000E+00   .00000E+00   .00000E+00 /
CC====== Fuel Consumption ==================================================/
CC   Thrust Specific Fuel Consumption Coefficients                          /
CD     .88100E+00   .16900E+05                                              /
CC   Descent Fuel Flow Coefficients                                        /
CD     .26805E+02   .45700E+05                                              /
CC   Cruise Corr.    unused       unused       unused       unused    /
CD     .10380E+01   .00000E+00   .00000E+00   .00000E+00   .00000E+00 /
CC====== Ground ============================================================/
CC        TOL          LDL         span         length       unused    /
CD     .23620E+04   .15550E+04   .44840E+02   .54080E+02   .00000E+00 /
CC=========================================================================/
FI                                                                          /
```

There are three types of lines in the OPF file with the line type identified by the first two characters in the line. These line types with their associated two leading characters are listed below.

CC	comment line
CD	data line
FI	end-of-file line

The comment lines are provided solely for the purpose of improving the readability of the file. All coefficients are contained within the CD lines in a fixed format. The end-of-file line is included as the last line in the file in order to facilitate the reading of the file in certain computing environments.

The data is organised into a total of eight blocks with each block separated by a comment line containing the block name and equal signs "=". These blocks are listed below and are described in further detail in the subsections below.

- file identification block
- aircraft type block
- mass block
- flight envelope block
- aerodynamics block
- engine thrust block,
- fuel consumption block
- ground movements block

6.4.1. File Identification Block

The file identification block provides information on the file name, creation date and modification date. The block consists of 11 comment lines. An example of the file identification block for the A306__.OPF file is shown below.

```
CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC A306__.OPF CCCCCCCCCCCCCC/
CC                                                                       /
CC                AIRCRAFT PERFORMANCE OPERATIONAL FILE                  /
CC                                                                       /
CC                                                                       /
CC      File_name: A306__.OPF                                            /
CC                                                                       /
CC      Creation_date: Mar 26 2002                                       /
CC                                                                       /
CC      Modification_date: Mar 26 2002                                   /
CC                                                                       /
```

The comment lines specify the file name along with the creation date and last modification date. The creation date indicates the date when the file was created for the first time. The modification date indicates when the contents of the file were last modified.

6.4.2. Aircraft Type Block

The OPF aircraft type block consists of 1 data line with 3 comment lines for a total of 4 lines. An example of the aircraft type block is given below.

```
        CC====== Actype ===================================================/
1 ->    CD   A306__         2 engines    Jet                    H          /
        CC   Airbus A300-B4-622 with PW4158  engines            wake       /
        CC                                                                 /
```

The data line specifies the following aircraft type parameters:

- ICAO aircraft code (followed by 2 or more underscore characters as required to form a six character string)
- number of engines, n_{eng}
- engine type
- wake category

The engine type can be one of the following three values: Jet, Turboprop or Piston. The wake category can be one of the four values J (jumbo), H (heavy), M (medium) or L (light).

The four values are specified in the following fixed format (Fortran notation)

'CD', 2X, A6, 10X, I1, 12X, A9, 17X, A1

The comment lines typically indicate the engine manufacturer's designation and the source of the performance coefficients.

6.4.3. Mass Block

The OPF mass block consists of 1 data line with 2 comment lines for a total of 3 lines.

An example of the mass block is given below.

```
        CC====== Mass (t) ================================================/
        CC    reference      minimum      maximum     max payload  mass grad  /
1 ->    CD     .14000E+03   .87000E+02   .17170E+03   .39000E+02   .14100E+00 /
```

The data line specifies the following BADA mass model parameters:

m_{ref} m_{min} m_{max} m_{pyld} G_w

These parameters are specified in the following fixed format (Fortran notation)

'CD', 2X, 5 (3X, E10.5)

6.4.4. Flight Envelope Block

The OPF flight envelope block consists of 1 data line with 2 comment lines for a total of 3 lines. An example of the flight envelope block is given below.

```
        CC====== Flight envelope ========================================/
        CC     VMO(KCAS)       MMO        Max.Alt       Hmax       temp grad  /
1 ->    CD     .33500E+03   .82000E+00   .41000E+05   .31600E+05  -.67000E+02 /
```

The date line specifies the following BADA speed envelope parameters:

V_{MO} M_{MO} h_{MO} h_{max} G_t

Note that all altitudes are expressed in feet.

These parameters are specified in the following fixed format (Fortran notation):

'CD', 2X, 5 (3X, E10.5)

6.4.5. Aerodynamics Block

The OPF aerodynamics block consists of 12 data lines and 8 comment lines for a total of 20 lines. An example of the aerodynamics block is given below.

```
        CC====== Aerodynamics ================================================/
        CC Wing Area and Buffet coefficients (SIM)                            /
        CCndrst Surf(m2)     Clbo(M=0)       k           CM16                 /
1 ->    CD 5   .26000E+03   .15300E+01   .10290E+01   .00000E+00              /
        CC   Configuration characteristics                                    /
        CC n Phase  Name    Vstall(KCAS)    CD0          CD2         unused   /
2 ->    CD 1 CR   Clean     .15100E+03   .19000E-01   .53000E-01   .00000E+00 /
3 ->    CD 2 IC   S15F00    .11700E+03   .33057E-01   .45362E-01   .00000E+00 /
4 ->    CD 3 TO   S15F00    .11700E+03   .33057E-01   .45362E-01   .00000E+00 /
5 ->    CD 4 AP   S15F15    .10900E+03   .38031E-01   .44932E-01   .00000E+00 /
6 ->    CD 5 LD   S30F40    .97000E+02   .78935E-01   .44822E-01   .00000E+00 /
        CC   Spoiler                                                          /
7 ->    CD 1      RET                                                         /
8 ->    CD 2      EXT                              .00000E+00   .00000E+00 /
        CC   Gear                                                             /
9 ->    CD 1      UP                                                          /
10 ->   CD 2      DOWN                .2250E-01    .00000E+00   .00000E+00 /
        CC   Brakes                                                           /
12 ->   CD 1      OFF                                                         /
13 ->   CD 2      ON                               .00000E+00   .00000E+00 /
```

The first data line specifies the following BADA aerodynamic model parameters:

S $C_{lbo(M=0)}$ k C_{M16}

These parameters are specified in the following fixed format (Fortran notation):

'CD', 2X, 4 (3X, E10.5)

Note that the "5" under the header "ndrst" stands for the five drag settings. Currently this is not used but is left in for compatibility requirements.

The next line holds besides the stall speed and flap setting for cruise as well as the values for the two drag coefficients for this configuration:

$(V_{stall})_{CR}$ C_{D0} C_{D2}

These parameters are specified in the following fixed format (Fortran notation):

'CD', 15X, 3 (3X, E10.5)

The next four data lines have the same format and correspond to the other configurations. The configurations are specified in the following order, corresponding to a semi-monotonically decreasing stall speed:

IC	initial climb
TO	take-off
AP	approach
LD	landing

The stall speed, $(V_{stall})_i$, is specified for each configuration, and C_{D0} and C_{D2} are given if available in the following fixed format (Fortran notation):

'CD', 15X, 3 (3X, E10.5)

In case the IC configuration is equal to the CR configuration, the values for C_{D0} and C_{D2} are mentioned only in the CR dataline. Note that C_{D0} and C_{D2} coefficients for IC and TO configurations are not used but are included for the reason of compatibility with previous versions.

The data lines 7 through 9 are not used but are included for the reason of compatibility with previous versions.

Dataline 10 holds the drag increment for landing gear down:

$C_{D0,\Delta LDG}$

The format of this line is:

'CD', 31X, E10.5

Datalines 11 and 12 are not used but are included for the reason of compatibility with previous versions.

6.4.6. Engine Thrust Block

The OPF engine thrust block consists of 3 data lines with 4 comment lines for a total of 7 lines. An example of the engine thrust block is given below.

```
        CC====== Engine Thrust ================================================/
        CC          Max climb thrust coefficients (SIM)                       /
1 ->    CD     .30400E+06   .44800E+05   .11600E-09   .67500E+01   .42600E-02 /
        CC      Desc(low)    Desc(high)   Desc level   Desc(app)    Desc(ld)  /
2 ->    CD     .73000E-02   .20600E-01   .80000E+04   .12000E+00   .36000E+00 /
        CC      Desc CAS     Desc Mach    unused       unused       unused    /
3 ->    CD     .28000E+03   .79000E+00   .00000E+00   .00000E+00   .00000E+00 /
```

The first data line specifies the following BADA parameters used to calculate the maximum climb thrust, that is:

$C_{Tc,1}$ $C_{Tc,2}$ $C_{Tc,3}$ $C_{Tc,4}$ $C_{Tc,5}$

These parameters are specified in the following fixed format (Fortran notation):

'CD', 2X, 5 (3X, E10.5)

The second data line specifies the following BADA parameters used to calculate cruise and descent thrust, that is:

$C_{Tdes,low}$ $C_{Tdes,high}$ $H_{p,des}$ $C_{Tdes,app}$ $C_{Tdes,ld}$

These parameters are specified in the following fixed format (Fortran notation):

'CD', 2X, 5 (3X, E10.5)

Note that the $C_{Tdes,app}$ and $C_{Tdes,ld}$ coefficients are determined in order to obtain a 3° descent gradient during approach and landing.

The third data line specifies the reference speeds during descent, that is:

$V_{des,ref}$ $M_{des,ref}$

These parameters are specified in the following fixed format (Fortran notation):

'CD', 2X, 2 (3X, E10.5)

Note that these two parameters are no longer used in BADA model implementation, but are left in place only to provide information on one of the reference speeds during descent used during the model identification.

The zero values at the end of this data line are not used but are included in the file due to compatibility requirements with previous versions.

6.4.7. Fuel Consumption Block

The OPF fuel consumption block consists of 3 data lines with 4 comment lines for a total of 7 lines. An example of a fuel consumption block is shown below.

```
        CC====== Fuel Consumption ============================================/
        CC   Thrust Specific Fuel Consumption Coefficients                    /
1 ->    CD     .88100E+00   .16900E+05                                       /
        CC   Descent Fuel Flow Coefficients                                   /
2 ->    CD     .26805E+02   .45700E+05                                       /
        CC   Cruise Corr.    unused       unused       unused       unused    /
3 ->    CD     .10380E+01   .00000E+00   .00000E+00   .00000E+00   .00000E+00 /
```

The first data line specifies the following BADA parameters for thrust specific fuel consumption.

C_{f1} C_{f2}

These parameters are specified in the following fixed format (Fortran notation):

'CD', 2X, 2 (3X, E10.5)

The second data line specifies the following BADA parameters for descent fuel flow.

C_{f3} C_{f4}

These parameters are specified in the following fixed format (Fortran notation):

'CD', 2X, 2 (3X, E10.5)

The third data line specifies the cruise fuel flow correction factor.

C_{fcr}

The parameter is specified in the following fixed format (Fortran notation):

'CD', 5X, E10.5

6.4.8. Ground Movement Block

The OPF ground movement block consists of 1 data line with 3 comment lines for a total of 4 lines. An example of a ground movement block is shown below. The ground movement block is the last block in the OPF file and is thus followed by the end-of-file line as shown below.

```
        CC====== Ground ======================================================/
        CC         TOL          LDL         span         length       unused    /
1 ->    CD     .23620E+04   .15550E+04   .44840E+02   .54080E+02   .00000E+00 /
        CC===================================================================/
        FI                                                                     /
```

The data line specifies the following BADA parameters for ground movements:

TOL LDL span length

These parameters are specified in the following fixed format (Fortran notation):

'CD', 2X, 4 (3X, E10.5)

6.5. APF FILE FORMAT

The Airlines Procedures File (APF) is an ASCII file which, for a particular aircraft type, specifies recommended speed procedures for climb, cruise, and descent conditions. An example of an APF file for the Airbus A306 aircraft is shown below.

```
CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC A306__.APF CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC/
CC                                                                                                           /
CC                 AIRLINES PROCEDURES FILE                                                                  /
CC                                                                                                           /
CC      File_name: A306__.APF                                                                                /
CC                                                                                                           /
CC      Creation_date: Mar 26 2002                                                                           /
CC                                                                                                           /
CC      Modification_date: Mar 26 2002                                                                       /
CC                                                                                                           /
CC                                                                                                           /
CC                                                                                                           /
CC   LO= 087.00 to ---.--  /  AV= ---.-- to ---.--  /  HI= ---.-- to 171.70                                  /
CC                                                                                                           /
CC===========================================================================================================/
CC  COM CO    Company name ------climb-------  --cruise--  -----descent------  --approach-  model- /
CC                         mass  lo  hi          lo  hi         hi  lo           (unused)         /
CC    version engines  ma  cas cas mc xxxx xx  cas cas mc  mc cas cas xxxx xx  xxx xxx xxx  opf___ /
CC===:=======:=======::==::===:===:==:====:==::===:===:==::==:===:===:====:==::===:===:===::======:/
CD  *** **    Default Company                                                                         /
CD    B4_622  PW4158   LO  250 300 79          250 310 79  79 280 250            0   0   0  A306__ /
CD    B4_622  PW4158   AV  250 300 79          250 310 79  79 280 250            0   0   0  A306__ /
CD    B4_622  PW4158C  HI  250 300 79          250 310 79  79 280 250            0   0   0  A306__ /
CC===:=======:=======::==::===:===:==:====:==::===:===:==::==:===:===:====:==::===:===:===::======:/
CC//////////////////////////////////  THE END  ////////////////////////////////////////////////////
```

There are two types of lines in the APF file with the line type identified by the first two characters in the line. These line types with their associated two leading characters are listed below:

CC - comment line
CD - data line

The last line in the file, as shown above, is also a comment line.

The comment lines are provided solely for the purpose of improving the readability of the file. All coefficients are contained within the CD lines in a fixed format.

The data is organised into 2 blocks separated by a comment line containing a string of equal signs, "=":

- file identification block
- speed procedures block

Each of the two blocks is described further in the subsections below.

6.5.1. File Identification Block

The file identification block provides information on the file name, creation date and modification date. The block consists of 14 comment lines. An example of a file identification block is shown below.

```
CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC A306__.APF CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC/
CC                                                                                                        /
CC                 AIRLINES PROCEDURES FILE                                                               /
CC                                                                                                        /
CC      File_name: A306__.APF                                                                             /
CC                                                                                                        /
CC      Creation_date: Mar 26 2002                                                                        /
CC                                                                                                        /
CC      Modification_date: Mar 26 2002                                                                    /
CC                                                                                                        /
CC                                                                                                        /
CC                                                                                                        /
CC   LO= 087.00 to ---.--  /  AV= ---.-- to ---.--  /  HI= ---.-- to 171.70                               /
CC                                                                                                        /
```

The comment lines provide background information on the file contents. In addition, the comment lines specify the file name along with the creation and last modification date. The creation date indicates the date when the file was created for the first time. The modification date indicates when the contents of the file were last modified.

The second last comment line in the identification block specifies three mass ranges for the aircraft in tonnes. That is, a low range (LO), average range (AV) and high range (HI). The definition of these ranges is used for interpreting the information presented below in the procedures specification block. In the example given above, all three ranges are assumed equivalent.

6.5.2. Procedures Specification Block

The APF procedures specification block consists of 4 data lines with 7 comment lines for a total of 11 lines. An example of a procedures specification block is shown below.

```
      CC=============================================================================================/
      CC  COM CO    Company name ------climb-------  --cruise--  -----descent------  --approach-  model- /
      CC                        mass  lo  hi             lo  hi         hi  lo           (unused)         /
      CC    version engines  ma  cas cas mc xxxx xx  cas cas mc  mc cas cas xxxx xx  xxx xxx xxx  opf___ /
      CC===:=======:=======::==::===:===:==:====:==::===:===:==::==:===:===:====:==::===:===:===::======:/
1 ->  CD  *** **    Default Company                                                                       /
2 ->  CD    B4_622  PW4158   LO  250 300 79          250 310 79  79 280 250            0   0   0  A306__ /
3 ->  CD    B4_622  PW4158   AV  250 300 79          250 310 79  79 280 250            0   0   0  A306__ /
4 ->  CD    B4_622  PW4158C  HI  250 300 79          250 310 79  79 280 250            0   0   0  A306__ /
      CC===:=======:=======::==::===:===:==:====:==::===:===:==::==:===:===:====:==::===:===:===::======:/
      CC//////////////////////////////////  THE END  ////////////////////////////////////////////////////
```

The first data line specifies the company name for which the next three datalines are valid. The company can be identified by its 3 and 2 letter code plus the company name. The dataline fomat is:

'CD', 2X, A3, 1X, A2, 4X, A15

As it is, within BADA all APF files specify procedures for only one "default" company.

The next three data lines specify the following parameters corresponding to climb, cruise and descent:

$V_{cl,1}$ $V_{cl,2}$ M_{cl} $V_{cr,1}$ $V_{cr,2}$ M_{cr} M_{des} $V_{des,2}$ $V_{des,1}$

Note that all Mach number values are also multiplied by a value of 100. For example, the 78 indicated for M_{cl} above corresponds to a Mach number of 0.78.

The three lines specify parameters for mass ranges of Low (LO), Average (AV) and High (HI) respectively. These parameters are specified in the following fixed format (Fortran notation):

'CD', 25X, 2(I3, 1X), I2, 10X, 2(I3, 1X), I2, 2X, I2, 2(1X, I3)

Note that approach values are set to zero. These values are not used but are included in the file due to compatibility requirements with previous versions.

Also, each line specifies an aircraft version number, engine, and operational model. The operational model is always the same as the file name. The version number may provide some additional information on the aircraft version covered by the file while the engine states which engine is used by the aircraft.

The file format is designed such that the four data lines can be repeated for the different companies which operate the aircraft and which may have different standard procedures. If data were to be provided for more than one company then the version, engine and operational model fields may be useful since different companies could operate different versions of the aircraft with different engines and thus different associated operational models.

6.6. PTF FILE FORMAT

The Performance Table File (PTF) is an ASCII file, which for a particular aircraft type specifies cruise, climb and descent performance at different flight levels. An example of a PTF file for the Airbus A306 aircraft is shown below.

```
                              BADA PERFORMANCE FILE         Apr 23 2002

AC/Type: A306
                              Source OPF File:               Mar 26 2002
                              Source APF file:               Mar 26 2002

 Speeds:   CAS(LO/HI)  Mach   Mass Levels [kg]         Temperature:   ISA
 climb   - 250/300     0.79   low     - 104400
 cruise  - 250/310     0.79   nominal - 140000         Max Alt. [ft]:  41000
 descent - 250/280     0.79   high    - 171700
=================================================================================
 FL |          CRUISE                |              CLIMB                |       DESCENT
    |  TAS          fuel              |  TAS       ROCD         fuel      |  TAS  ROCD    fuel
    | [kts]       [kg/min]            | [kts]      [fpm]       [kg/min]   | [kts] [fpm] [kg/min]
    |          lo   nom    hi         |          lo  nom   hi     nom       |        nom    nom
=================================================================================
  0 |                                 |  157   2210 1990 1620   270.3     |  131   760    97.2
    |                                 |
  5 |                                 |  158   2190 1970 1600   267.3     |  132   780    96.1
    |                                 |
 10 |                                 |  159   2170 1950 1570   264.3     |  138   800    95.0
    |                                 |
 15 |                                 |  166   2290 2030 1650   261.5     |  149   850    94.0
    |                                 |
 20 |                                 |  167   2270 2010 1620   258.5     |  181  1020    31.0
    |                                 |
 30 |  230    61.2  81.4 104.3        |  190   2750 2360 1920   253.0     |  230  1360    25.0
    |                                 |
 40 |  233    61.2  81.4 104.4        |  225   3350 2780 2270   247.7     |  233  1380    24.5
    |                                 |
 60 |  272    65.9  81.7  99.6        |  272   4210 3070 2370   236.8     |  240  1410    23.3
    |                                 |
 80 |  280    65.8  81.7  99.7        |  280   4040 2930 2230   225.7     |  280  1550    22.1
    |                                 |
100 |  289    65.8  81.7  99.8        |  289   3860 2780 2090   214.8     |  289  1590    20.9
    |                                 |
120 |  297    65.7  81.7  99.8        |  356   3820 2800 2170   204.8     |  332  1880    19.8
    |                                 |
140 |  306    65.6  81.7  99.9        |  366   3590 2610 2000   194.3     |  342  1920    18.6
    |                                 |
160 |  389    82.4  93.1 105.3        |  377   3360 2410 1820   184.1     |  353  1960    17.4
    |                                 |
180 |  401    82.1  92.9 105.1        |  388   3120 2220 1650   174.2     |  363  2000    16.2
    |                                 |
200 |  413    81.7  92.6 104.9        |  400   2880 2020 1470   164.5     |  375  2040    15.1
    |                                 |
220 |  425    81.3  92.3 104.7        |  412   2630 1810 1290   155.0     |  386  2080    13.9
    |                                 |
240 |  438    80.9  91.9 104.5        |  425   2380 1610 1100   145.8     |  398  2120    12.7
    |                                 |
260 |  452    80.4  91.6 104.3        |  438   2130 1400  920   136.9     |  411  2160    11.6
    |                                 |
280 |  466    79.9  91.2 104.1        |  452   1880 1200  730   128.1     |  424  2200    10.4
    |                                 |
290 |  468    78.4  90.1 103.4        |  459   1760 1090  640   123.9     |  431  2220     9.8
    |                                 |
310 |  464    74.3  87.0 101.5        |  464   2200 1290  660   115.4     |  444  2250     8.6
    |                                 |
330 |  459    70.6  84.7 100.6        |  459   1950 1050  420   107.2     |  459  2290     7.4
    |                                 |
350 |  455    67.6  83.0  97.9        |  455   1700  810  170    99.2     |  455  3150     6.3
    |                                 |
370 |  453    65.1  82.0  90.3        |  453   1320  510    0    91.6     |  453  2850     5.1
    |                                 |
390 |  453    63.2  81.9  83.0        |  453   1080  260    0    84.1     |  453  2850     3.9
    |                                 |
410 |  453    61.9  75.9  75.9        |  453    830   10    0    77.0     |  453  2880     2.8
=================================================================================
```

The OPF and APF files are generated as a result of a modelling process using MatLab [RD6]. Once these two files are generated, the PTF can be automatically generated. A brief summary of the format of these files is given below.

The header of each PTF file contains information as described below.

file creation date: This is in the first line, at the top-right corner

aircraft type: This is in the third line.

source file dates: The last modification dates of the OPF and APF files which were used to create the PTF file are given in the 4th and 5th lines respectively.

Speeds: The speed laws for climb, cruise and descent are specified in lines 8, 9 and 10, that is:

climb	$\min(V_{cl,1}, 250kt) / V_{cl,2}$	M_{cl}
cruise	$\min(V_{cr,1}, 250kt) / V_{cr,2}$	M_{cr}
descent	$\min(V_{des,1}, 250kt) / V_{des,2}$	M_{des}

Mass levels: The performance tables provide data for three different mass levels in lines 8, 9 and 10, that is:

low	$1.2\ m_{min.}$
nominal	m_{ref}
high	m_{max}

Note that the low mass is not the minimum mass but 1.2 times the minimum mass.

Temperature: The temperature is mentioned in line 7. All PTF files currently provide data for ISA conditions only.

Maximum altitude: The maximum altitude as specified in the OPF file, h_{MO}, is given in line 9.

The table of performance data within the file consists of 13 columns. Each of these columns is described below:

Column 1	FL
Column 2	cruise TAS (nominal mass) [knots]
Column 3	cruise fuel consumption (low mass) [kg/min]
Column 4	cruise fuel consumption (nominal mass) [kg/min]
Column 5	cruise fuel consumption (high mass) [kg/min]
Column 6	climb TAS (nominal mass) [knots]
Column 7	rate of climb with reduced power (low mass) [ft/min]
Column 8	rate of climb with reduced power (nominal mass) [ft/min]
Column 9	rate of climb with reduced power (high mass) [ft/min]
Column 10	climb fuel consumption (nominal mass) [kg/min]
Column 11	descent TAS (nominal mass) [knots]
Column 12	rate of descent (nominal mass) [ft/min]
Column 13	descent fuel consumption (nominal mass) [kg/min]

The format for data presented in each line of the table is as follows (Fortran notation):

I3, 4X, I3, 2X, 3(1X, F5.1), 5X, I3, 2X, 3(1X, I5), 3X, F5.1, 5X, I3, 2X, I5, 2X, F5.1

Further explanatory notes on the data presented in the performance tables are given below:

(a) Cruise data is only specified for flight levels greater than or equal to 30.

(b) Performance data is specified up to a maximum flight level of 510 or to highest level for which a positive rate of climb can be achieved at the low mass.

(c) True airspeed for climb, cruise and descent is determined based on the speed schedules specified in Sections 4.1, 4.2 and 4.3 respectively.

(d) Rates of climb are calculated at each flight level assuming the energy share factors associated with constant CAS or constant Mach speed laws and using the reduced power correction as given in Section 3.8.

(e) The fuel consumption in climb is independent of the aircraft mass and thus only one value is given. There are three different climb rates however corresponding to low, nominal and high mass conditions.

(f) The rate of descent and fuel consumption in descent is calculated assuming the nominal mass. Values for other mass conditions are not given.

(g) Discontinuities in climb rate can occur for the following reasons:
- change in speed between flight levels (e.g. removal of 250 kt restriction above FL100),
- transition from constant CAS to constant Mach (typically around FL300),
- transition through the tropopause (FL360 for ISA),
- end of the application scope for reduced climb power (at 80% of h_{max}).

(h) Discontinuities in descent rate can occur for the following reasons:
- transition through tropopause (FL360 for ISA),
- transition from constant Mach to constant CAS,
- change in assumed descent thrust (specified by the BADA h_{des} parameter),
- change to approach or landing aerodynamic configuration,
- change in speed between flight levels (e.g. application of 250 kt limit below FL100).

(i) The PTF files are made with "non-clean" configuration data for approach and landing when such data is available (see Section 3.6.1).

(j) The performance data presented in the table are computed by using ‘point type’ calculation, that is without performing integration over time: aircraft weight is constant and does not account for consumed fuel, and speed changes take place immediately.

(k) The flight envelope limitations are not taken into account for calculation of performance parameters[8].

Note that all PTF files are available in document form in [RD9].

[8] Example: cruise fuel flow is calculated without checking, for given aircraft weight, speed and FL, that aircraft drag is lower than maximum available cruise thrust.

6.7. PTD FILE FORMAT

In addition to the data provided in the PTF file, more detailed climb and descent performance data are presented in the PTD file. An example of a PTD file for the Airbus A306 aircraft is shown below (partial listing):

```
Low mass CLIMBS
===============

 FL[-] T[K] p[Pa] rho[kg/m3] a[m/s] TAS[kt] CAS[kt]  M[-] mass[kg] Thrust[N] Drag[N] Fuel[kgm] ESF[-] ROC[fpm] TDC[N]  PWC[-]
     0 288 101325    1.225    340   136.35  136.35 0.21  104400   297160   85670   215.8   0.98    2452  186284   0.88
     5 287  99508    1.207    340   137.34  136.35 0.21  104400   294268   85680   213.9   0.98    2435  183727   0.88
    10 286  97717    1.190    339   138.34  136.35 0.21  104400   291385   85691   212.0   0.98    2417  181179   0.88
    15 285  95952    1.172    339   144.45  141.35 0.22  104400   288510   82072   211.0   0.97    2527  181833   0.88
    20 284  94213    1.155    338   145.51  141.35 0.22  104400   285643   82082   209.1   0.97    2509  179299   0.88
    30 282  90812    1.121    337   168.52  161.35 0.26  104400   279935   72295   209.0   0.96    2937  182892   0.88
    40 280  87511    1.088    336   202.72  191.35 0.31  104400   274260   67093   210.7   0.95    3470  182476   0.88
    60 276  81200    1.024    333   272.30  250.00 0.42  104400   263011   74643   213.7   0.91    4075  165917   0.88
    80 272  75262    0.963    331   280.34  250.00 0.44  104400   251895   74535   206.0   0.91    3925  156222   0.88
   100 268  69682    0.905    328   345.37  300.00 0.54  104400   240914   91120   207.0   0.87    3905  131941   0.88
   120 264  64441    0.849    326   355.51  300.00 0.56  104400   230066   90785   199.1   0.86    3703  122681   0.88
   140 260  59524    0.796    324   366.04  300.00 0.58  104400   219352   90425   191.3   0.85    3495  113561   0.88
   160 256  54915    0.746    321   376.97  300.00 0.60  104400   208772   90038   183.6   0.84    3281  104583   0.88
   180 252  50600    0.698    319   388.32  300.00 0.63  104400   198326   89622   175.8   0.83    3060   95748   0.88
   200 249  46563    0.653    316   400.10  300.00 0.65  104400   188013   89175   168.1   0.82    2834   87059   0.88
   220 245  42791    0.610    314   412.32  300.00 0.68  104400   177835   88694   160.4   0.81    2601   78516   0.88
   240 241  39271    0.569    311   425.00  300.00 0.70  104400   167790   88179   152.7   0.80    2364   70123   0.88
   260 237  35989    0.530    308   438.16  300.00 0.73  104400   157879   87627   145.0   0.79    2122   61879   0.88
   280 233  32932    0.493    306   451.80  300.00 0.76  104400   148102   87036   137.3   0.78    1875   53788   0.88
   290 231  31485    0.475    304   458.81  300.00 0.78  104400   143263   86725   133.4   0.78    1749   49800   0.88
   310 227  28745    0.442    302   463.54  293.28 0.79  104400   133687   83916   124.9   1.09    2184   43839   0.88
   330 223  26201    0.410    299   459.48  280.58 0.79  104400   124245   79587   115.8   1.09    2205   44658   1.00
   350 219  23842    0.380    297   455.37  268.17 0.79  104400   114936   75881   106.8   1.09    1911   39055   1.00
   370 217  21663    0.348    295   453.12  256.08 0.79  104400   105761   72808    98.1   1.00    1470   32953   1.00
   390 217  19677    0.316    295   453.12  244.46 0.79  104400    96720   70398    89.7   1.00    1174   26322   1.00
   410 217  17874    0.287    295   453.12  233.34 0.79  104400    87813   68640    81.5   1.00     855   19173   1.00
```

The performance values presented in the PTD file are a superset of the climb and descent performance values presented in the PTF file. They are generated in the same conditions as the corresponding PTF file: same aircraft, same source OPF and APF files, same speed laws, same mass levels, same temperature and same flight levels. The purpose of this file is mainly to provide the user with a greater number of computed parameters, especially intermediate parameters used to compute the final TAS and ROCD, which may be useful to validate an implementation of the BADA model.

The files contains performance data consisting of 4 sections:

- low mass climb performance
- nominal mass climb performance
- high mass climb performance
- nominal mass descent performance

Each section contains a table that presents, for several flight levels, a set of performance parameters spread across 16 columns. Each of these columns is described below:

Column 1	Flight level [FL]
Column 2	Temperature [K]
Column 3	Pressure [Pa]
Column 4	Air density [kg/m^3]
Column 5	Speed of sound [m/s]
Column 6	TAS [kt]
Column 7	CAS [kt]
Column 8	Mach [dimensionless]
Column 9	Mass [kg]
Column 10	Thrust [N]
Column 11	Drag [N]
Column 12	Fuel flow [kg/min]
Column 13	Energy share factor [dimensionless]
Column 14	Rate of climb/descent [ft/min]
Column 15	(Thr – D) · $C_{pow,red}$ [kg/min] (see section 3.8)
Column 16	- climb tables: Power reduction coefficient $C_{pow,red}$ [dimensionless] - descent table: Descent gradient [degree]

The format for data presented in each line of the table is as follows (Fortran notation):

Climb tables:

I6, 1X, I3, 1X, I6, 1X, F7.3, 1X, I7, 2(1X, F8.2), 1X, F7.2, 1X, I6, 2(1X, I9), 1X, F7.1, 1X, F7.2, 1X, I7, 1X, I8, 1X, F7.2

Descent tables:

I6, 1X, I3, 1X, I6, 1X, F7.3, 1X, I7, 2(1X, F8.2), 1X, F7.2, 1X, I6, 2(1X, I9), 1X, F7.1, 1X, F7.2, 1X, I7, 1X, I8, 1X, F8.2

6.8. BADA.GPF FILE FORMAT

The BADA.GPF file is an ASCII file which specifies the values of the global aircraft parameters (see Section 5). The complete BADA.GPF file is shown below:

```
CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC BADA.GPF CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC/
CC                                                                              /
CC                GLOBAL PARAMETERS FILE                                        /
CC                                                                              /
CC     File_name: BADA.GPF                                                      /
CC                                                                              /
CC     Creation_date: Mar 26 2002                                               /
CC                                                                              /
CC     Modification_date: Mar 26 2002                                           /
CC                                                                              /
CC                                                                              /
CC======== Class ================================================================/
CC                                                                              /
CC Flight = civ,mil                                                             /
CC Engine = jet,turbo,piston                                                    /
CC Phase  = to,ic,cl,cr,des,hold,app,lnd,gnd                                    /
CC                                                                              /
CC======== Parameters List ======================================================/
CC                                                                              /
CC Name             Unit                                                        /
CC Parameter       Flight  Engine           Phase                        Value  /
CC                                                                              /
CC max. long. acc.  [fps2]                                                      /
CD acc_long_max    civ     jet,turbo,piston to,ic,cl,cr,des,hold,app,lnd  .20000E+01 /
CC max. norm. acc.  [fps2]                                                      /
CD acc_norm_max    civ     jet,turbo,piston to,ic,cl,cr,des,hold,app,lnd  .50000E+01 /
CC nom. bank angle  [deg]                                                       /
CD ang_bank_nom    civ     jet,turbo,piston to,lnd                        .15000E+02 /
CC nom. bank angle  [deg]                                                       /
CD ang_bank_nom    civ     jet,turbo,piston ic,cl,cr,des,hold,app         .35000E+02 /
CC nom. bank angle  [deg]                                                       /
CD ang_bank_nom    mil     jet,turbo,piston to,ic,cl,cr,des,hold,app,lnd  .50000E+02 /
CC max. bank angle  [deg]                                                       /
CD ang_bank_max    civ     jet,turbo,piston to,lnd                        .25000E+02 /
CC max. bank angle  [deg]                                                       /
CD ang_bank_max    civ     jet,turbo,piston hold                          .35000E+02 /
CC max. bank angle  [deg]                                                       /
CD ang_bank_max    civ     jet,turbo,piston ic,cl,cr,des,app              .45000E+02 /
CC max. bank angle  [deg]                                                       /
CD ang_bank_max    mil     jet,turbo,piston to,ic,cl,cr,des,hold,app,lnd  .70000E+02 /
CC exp. desc. fact. [-]                                                         /
CD C_des_exp       civ,mil jet,turbo,piston des                           .16000E+01 /
CC to thrust factor [-]                                                         /
CD C_th_to         mil,civ jet,turbo,piston to                            .12000E+01 /
CC cr thrust factor [-]                                                         /
CD C_th_cr         mil,civ jet,turbo,piston cr                            .95000E+00 /
CC max alt for to  [ft]                                                         /
CD H_max_to        mil,civ jet,turbo,piston to                            .40000E+03 /
CC max alt for ic  [ft]                                                         /
CD H_max_ic        mil,civ jet,turbo,piston ic                            .20000E+04 /
CC max alt for app  [ft]                                                        /
CD H_max_app       mil,civ jet,turbo,piston app                           .80000E+04 /
CC max alt for ld  [ft]                                                         /
CD H_max_ld        mil,civ jet,turbo,piston lnd                           .30000E+04 /
CC min speed coef.  [-]                                                         /
CD C_v_min         mil,civ jet,turbo,piston cr,ic,cl,des,hold,app,lnd     .13000E+01 /
CC min speed coef.  [-]                                                         /
CD C_v_min_to      mil,civ jet,turbo,piston to                            .12000E+01 /
CC spd incr FL < 15 [KCAS]                                                      /
CD V_cl_1          mil,civ jet             cl                             .50000E+01 /
CC spd incr FL < 30 [KCAS]                                                      /
CD V_cl_2          mil,civ jet             cl                             .10000E+02 /
CC spd incr FL < 40 [KCAS]                                                      /
CD V_cl_3          mil,civ jet             cl                             .30000E+02 /
CC spd incr FL < 50 [KCAS]                                                      /
CD V_cl_4          mil,civ jet             cl                             .60000E+02 /
CC spd incr FL < 60 [KCAS]                                                      /
CD V_cl_5          mil,civ jet             cl                             .80000E+02 /
```

```
CC spd incr FL < 5  [KCAS]                                                    /
CD V_cl_6          mil,civ turbo,piston     cl                      .20000E+02 /
CC spd incr FL < 10 [KCAS]                                                    /
CD V_cl_7          mil,civ turbo,piston     cl                      .30000E+02 /
CC spd incr FL < 15 [KCAS]                                                    /
CD V_cl_8          mil,civ turbo,piston     cl                      .35000E+02 /
CC spd incr FL < 10 [KCAS]                                                    /
CD V_des_1         mil,civ jet,turbo        des                     .50000E+01 /
CC spd incr FL < 15 [KCAS]                                                    /
CD V_des_2         mil,civ jet,turbo        des                     .10000E+02 /
CC spd incr FL < 20 [KCAS]                                                    /
CD V_des_3         mil,civ jet,turbo        des                     .20000E+02 /
CC spd incr FL < 30 [KCAS]                                                    /
CD V_des_4         mil,civ jet,turbo        des                     .50000E+02 /
CC spd incr FL < 5  [KCAS]                                                    /
CD V_des_5         mil,civ piston           des                     .50000E+01 /
CC spd incr FL < 10 [KCAS]                                                    /
CD V_des_6         mil,civ piston           des                     .10000E+02 /
CC spd incr FL < 15 [KCAS]                                                    /
CD V_des_7         mil,civ piston           des                     .20000E+02 /
CC hold. spd FL < 140 [KCAS]                                                  /
CD V_hold_1        mil,civ jet,turbo,piston hold                    .23000E+03 /
CC hold. spd FL < 200 [KCAS]                                                  /
CD V_hold_2        mil,civ jet,turbo,piston hold                    .24000E+03 /
CC hold. spd FL < 340 [KCAS]                                                  /
CD V_hold_3        mil,civ jet,turbo,piston hold                    .26500E+03 /
CC hold. spd FL > 340 [M]                                                     /
CD V_hold_4        mil,civ jet,turbo,piston hold                    .83000E+00 /
CC backtrack spd    [KCAS]                                                    /
CD V_backtrack     mil,civ jet,turbo,piston gnd                     .35000E+02 /
CC taxi spd         [KCAS]                                                    /
CD V_taxi          mil,civ jet,turbo,piston gnd                     .15000E+02 /
CC apron spd        [KCAS]                                                    /
CD V_apron         mil,civ jet,turbo,piston gnd                     .10000E+02 /
CC gate spd         [KCAS]                                                    /
 CD V_gate        mil,civ jet,turbo,piston gnd                    .50000E+01 /
CC Piston pow. red. [-]                                                       /
CD C_red_piston    mil,civ piston           ic,cl                   .000000+00 /
CC Turbo pow. red.  [-]                                                       /
CD C_red_turbo     mil,civ turbo            ic,cl                   .250000+00 /
CC Jet power red.   [-]                                                       /
CD C_red_jet       mil,civ jet              ic,cl                   .150000+00 /
FI============================================================================/
CC//////////////////////////////  THE END  ////////////////////////////////////////
```

There are three types of lines in the BADA.GPF file with the line type identified by the first two characters in the line. These line types with their associated two leading characters are listed below.

CC	comment line
CD	data line
FI	end-of-file line

The data is organised into three blocks separated by a comment line consisting of the block name and equal signs "=":

- file identification block
- class block
- parameter block

Each of these blocks is described in the subsections below.

6.8.1. File Identification Block

The file identification block provides information on the file name, creation and modification date. The block consists of 12 comment lines.

```
CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC BADA.GPF CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC/
CC                                                                                        /
CC                GLOBAL PARAMETERS FILE                                                  /
CC                                                                                        /
CC     File_name: BADA.GPF                                                                /
CC                                                                                        /
CC     Creation_date: Mar 26 2002                                                         /
CC                                                                                        /
CC     Modification_date: Mar 26 2002                                                     /
```

The comment lines specify the file name along with the creation date and last modification date. The creation date indicates the date when the file was created for the first time. The modification date indicates when the contents of the file were last modified.

6.8.2. Class Block

The class block consists of 6 comment lines and defines the three classes (Flight, Engine and Phase) and their instances that are used in the BADA.GPF file.

```
CC======== Class ============================================================================/
CC                                                                                          /
CC Flight = civ,mil                                                                         /
CC Engine = jet,turbo,piston                                                                /
CC Phase  = to,ic,cl,cr,des,hold,app,lnd,gnd                                                /
CC                                                                                          /
```

With:

civ	=	civil flight
mil	=	military flight
jet	=	jet engine
turbo	=	turboprop engine
piston	=	piston engine
to	=	take-off
ic	=	initial climb
cl	=	climb
cr	=	cruise
des	=	descent
hold	=	holding
app	=	approach
lnd	=	landing
gnd	=	ground

6.8.3. Parameter Block

The parameter block contains the values of the global aircraft parameters. This block has 5 comment lines plus a comment line and a dataline for each parameter.

```
        CC======== Parameters List ============================================================/
        CC                                                                                     /
        CC Name             Unit                                                               /
        CC Parameter       Flight  Engine           Phase                         Value       /
        CC                                                                                     /
        CC max. long. acc.  [fps2]                                                             /
1 ->    CD acc_long_max    civ     jet,turbo,piston to,ic,cl,cr,des,hold,app,lnd  .20000E+01 /
        CC max. norm. acc.  [fps2]                                                             /
2 ->    CD acc_norm_max    civ     jet,turbo,piston to,ic,cl,cr,des,hold,app,lnd  .50000E+01 /
        CC nom. bank angle  [deg]                                                              /
3 ->    CD ang_bank_nom    civ     jet,turbo,piston to,lnd                        .15000E+02 /
```

The parameter comment line contains the parameter name and its unit.

The parameter data line contains five fields:

(a) Parameter Field: This field identifies the parameter.

(b) Flight Field: This field identifies whether the parameter is valid for a civil flight, a military flight or both.

(c) Engine Field: This field identifies the engine type (jet, turboprop or piston) for which the parameter is valid.

(d) Phase Field: This field identifies for which flight phase the parameter is valid. 8 different flight phases are currently defined

(e) Value Field: The value field gives the value of the parameter.

The fields above are specified in the following fixed format (Fortran notation):

'CD', 1X, A15, 1X, A7, 1X, A16, 1X, A29, 1X, E10.5

The parameter list continues until 'FI' (end of file) is reached.

7. REMOTE FILE ACCESS

The files associated with BADA Revision 3.11 are accessible through the BADA Support Application (BSA). The BSA is a Web application that provides BADA users with the ability to exchange requests, as well as data files and documents, with the BADA team members. It is also used for data repository of the BADA release files and documents.

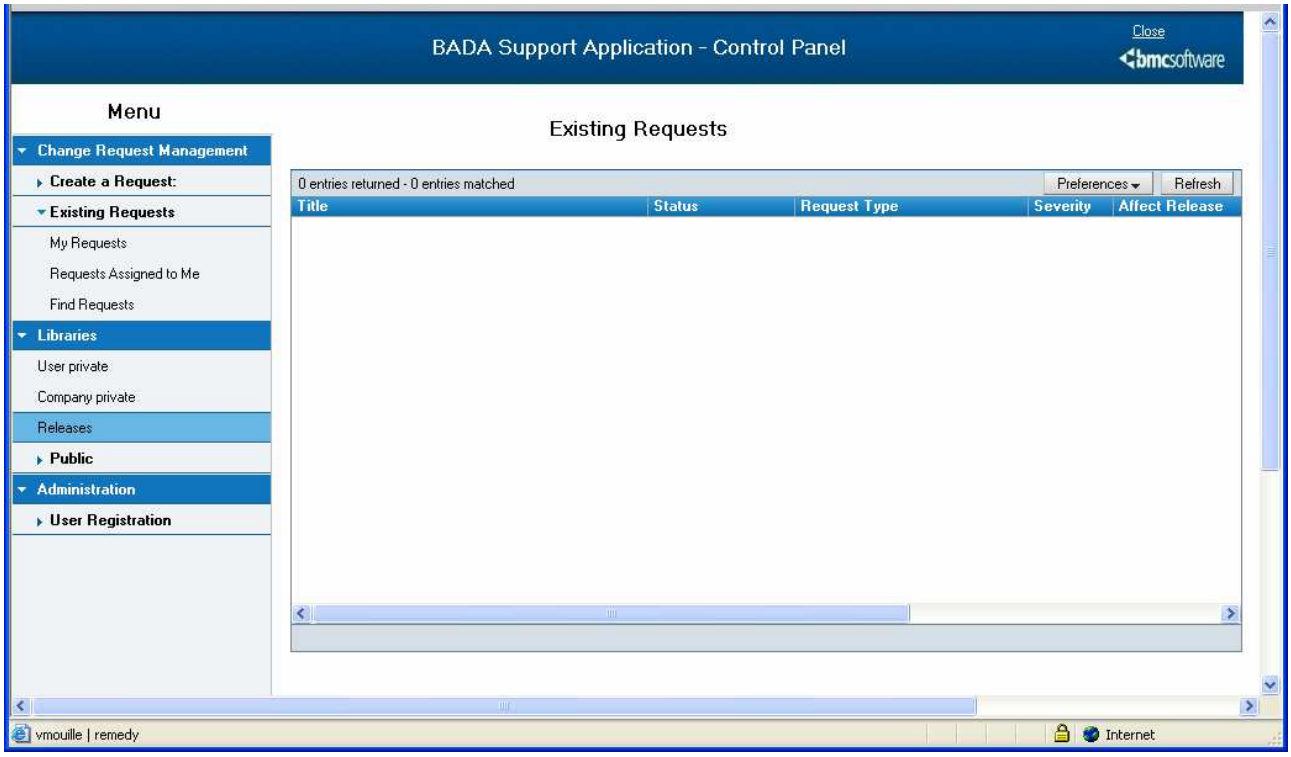
The right to use the BSA is granted to the licensed user of BADA. The application can be accessed by using a dedicated login and password as provided by the BADA team.

Once granted the access right to BSA, the user can access the application at this address:

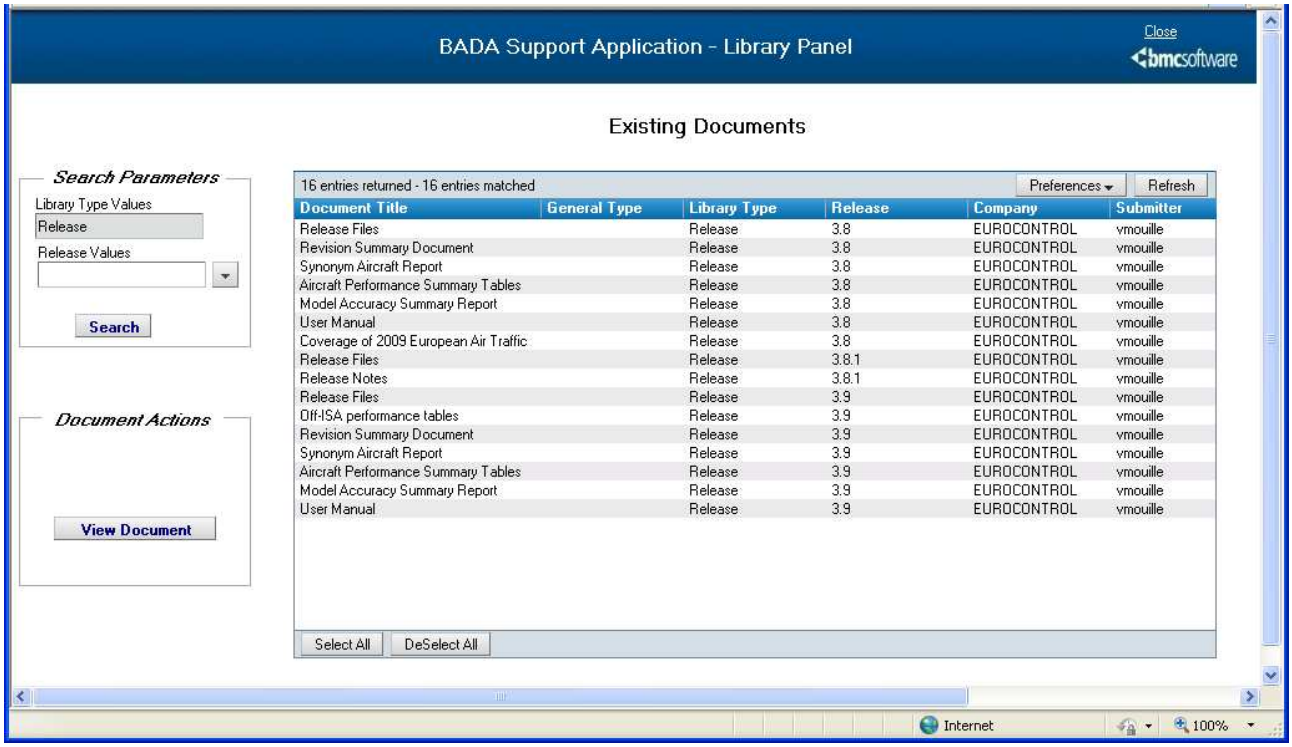
https://remedyweb.eurocontrol.fr

by using the BADA Support Application link and logging in with the provided login/password.

Once logged in, the user can access BADA releases through the Librairies→Releases item located in the main menu:



Then the release library page opens up, from where the user can download the BADA release files:



This process, as well as the general usage of the BSA application, is described in detail in [RD11].

Note that enquiries can be addressed to the following addresses:

E-mail: eec.bada@eurocontrol.int

Fax: + 33 1 69 88 73 33

BADA web page: http://www.eurocontrol.int/services/bada

APPENDIX A

BADA REVISION 3.11 – LIST OF AVAILABLE AIRCRAFT MODELS

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Table 7-1: List of Aircraft Types Supported by BADA Revision 3.11

A/C Code	Model Type	Aircraft manufacturer	Aircraft model	Synonym aircraft	h_{MO} [ft]	h_{max} [ft]	WTC
A10	equiv.	FAIRCHILD	THUNDERBOLT II	FGTN	50000		M
A124	direct	ANTONOV	AN-124 RUSLAN	A124	39370	30583	H
A140	direct	ANTONOV	AN-140	A140	24934	20296	M
A148	direct	ANTONOV	AN-148-100	A148	40020	38152	M
A306	direct	AIRBUS	A300B4-600	A306	41000	32378	H
A30B	direct	AIRBUS	A300B4-200	A30B	39000	31966	H
A310	direct	AIRBUS	A310	A310	41000	35718	H
A318	direct	AIRBUS	A318	A318	41000	37606	M
A319	direct	AIRBUS	A319	A319	41000	36365	M
A320	direct	AIRBUS	A320	A320	41000	33295	M
A321	direct	AIRBUS	A321	A321	39100	35396	M
A332	direct	AIRBUS	A330-200	A332	41000	36210	H
A333	direct	AIRBUS	A330-300	A333	41000	36392	H
A342	direct	AIRBUS	A340-200	A342	41500	31369	H
A343	direct	AIRBUS	A340-300	A343	41500	31059	H
A345	direct	AIRBUS	A340-500	A345	41450	32862	H
A346	direct	AIRBUS	A340-600	A346	41500	33613	H
A388	direct	AIRBUS	A380-800	A388	43100	34330	J
A3ST	direct	AIRBUS	A-300ST Beluga	A3ST	35000		H
A4	equiv.	DOUGLAS	SUPER SKYHAWK	FGTN	50000		M
A400	equiv.	AIRBUS	A-400M	A3ST	35000		H
A6	equiv.	GRUMMAN	INTRUDER	FGTN	50000		M
A7	equiv.	VOUGHT	CORSAIR II A7	FGTN	50000		M
A748	equiv.	AVRO	AVRO 748	ATP	25000	21628	M
AA5	equiv.	GULFSTREAM	Cheetah AA-5	P28A	12000		L
AC11	equiv.	ROCKWELL	COMMANDER 112	SR22	17500		L
AC90	equiv.	ROCKWELL	TURBO COMMANDER 690B	PAY3	33000		L
AC95	equiv.	GULFSTREAM	Jetprop Commander 980	PAY3	33000		L
AEST	equiv.	TED SMITH	AEROSTAR	MU2	28000		L
AFOX	equiv.	HALLEY	APOLO FOX	P28A	12000		L

A/C Code	Model Type	Aircraft manufacturer	Aircraft model	Synonym aircraft	h_{MO} [ft]	h_{max} [ft]	WTC
AJET	equiv.	DASSAULT	ALPHA JET	FGTN	50000		M
ALSL	equiv.	AIRLONY	SKYLANE	P28A	12000		L
AMX	equiv.	EMBRAER	AMX	FGTN	50000		M
AN12	equiv.	ANTONOV	AN-12	C130	33000	21891	M
AN24	direct	ANTONOV	AN-24	AN24	27560	24490	M
AN26	equiv.	ANTONOV	AN-26	AN30	29528	21982	M
AN28	direct	ANTONOV	AN-28	AN28	13780		L
AN30	direct	ANTONOV	AN-30	AN30	29528	21982	M
AN32	direct	ANTONOV	AN-32B	AN32	30840	29601	M
AN38	direct	ANTONOV	AN-38-100	AN38	13780		M
AN72	equiv.	ANTONOV	AN-72	F28	35000	31000	M
APM4	equiv.	ISSOIRE	APM 40 SIMBA	P28A	12000		L
ASTR	equiv.	IAI	1125 Astra	FA10	45000	38400	M
AT43	direct	ATR	ATR 42-300	AT43	25000	22699	M
AT44	equiv.	ATR	ATR 42-400	AT45	25000	23591	M
AT45	direct	ATR	ATR 42-500	AT45	25000	23591	M
AT46	equiv.	ATR	ATR 42-600	AT45	25000	23591	M
AT72	direct	ATR	ATR 72-200	AT72	25000	20317	M
AT73	direct	ATR	ATR 72-210	AT73	25000	20943	M
AT75	direct	ATR	ATR 72-500	AT75	25000	20779	M
AT76	equiv.	ATR	ATR 72-600	AT75	25000	20779	M
ATLA	equiv.	DASSAULT	1150 ATLANTIC	E120	32000		M
ATP	direct	BAE	ADVANCED TURBOPROP	ATP	25000	21628	M
B1	equiv.	ROCKWELL	B1 LANCER	FGTL	41000		H
B190	equiv.	BEECH	1900	JS32	25000		L
B350	equiv.	BEECH	SUPER KING AIR 350	BE20	35000		L
B461	equiv.	BAE	146-100/RJ	B462	31000	32716	M
B462	direct	BAE	146-200/RJ	B462	31000	32716	M
B463	direct	BAE	146-300/RJ	B463	31000	29305	M
B52	equiv.	BOEING	B-52 Stratofortress	FGTL	41000		H
B701	equiv.	BOEING	707-100	B752	42000	35478	M

A/C Code	Model Type	Aircraft manufacturer	Aircraft model	Synonym aircraft	h_{MO} [ft]	h_{max} [ft]	WTC
B703	direct	BOEING	707-300	B703	42000	35000	H
B712	direct	BOEING	717-200	B712	37000	35187	M
B720	equiv.	BOEING	B720B	B752	42000	35478	M
B721	equiv.	BOEING	727-100	B752	42000	35478	M
B722	direct	BOEING	727-200	B722	37000	33845	M
B731	equiv.	BOEING	737-100	T134	39000	34764	M
B732	direct	BOEING	737-200	B732	37000	34508	M
B733	direct	BOEING	737-300	B733	39000	33636	M
B734	direct	BOEING	737-400	B734	37000	33448	M
B735	direct	BOEING	737-500	B735	37000	34768	M
B736	direct	BOEING	737-600	B736	41000	39276	M
B737	direct	BOEING	737-700	B737	41000	37332	M
B738	direct	BOEING	737-800	B738	41000	34982	M
B739	direct	BOEING	737-900	B739	41000	34683	M
B741	equiv.	BOEING	747-100	B743	45000	30943	H
B742	direct	BOEING	747-200	B742	45000	33180	H
B743	direct	BOEING	747-300	B743	45000	30943	H
B744	direct	BOEING	747-400	B744	45000	32726	H
B748	direct	BOEING	747-8	B748	42100	32973	H
B74D	equiv.	BOEING	747-400 DOMESTIC	B744	45000	32726	H
B74R	equiv.	BOEING	747SR	B743	45000	30943	H
B74S	equiv.	BOEING	747-SP	B742	45000	33180	H
B752	direct	BOEING	757-200	B752	42000	35478	M
B753	direct	BOEING	757-300	B753	43000	33339	M
B762	direct	BOEING	767-200	B762	43000	35861	H
B763	direct	BOEING	767-300	B763	43100	36502	H
B764	direct	BOEING	767-400	B764	45000	33210	H
B772	direct	BOEING	777-200 ER	B772	43100	34643	H
B773	direct	BOEING	777-300	B773	43100	31857	H
B77L	direct	BOEING	777-200 LRF	B77L	43100	35104	H
B77W	direct	BOEING	777-300 ER	B77W	43100	34314	H
B788	direct	BOEING	787-8	B788	43100	35901	H

A/C Code	Model Type	Aircraft manufacturer	Aircraft model	Synonym aircraft	h_{MO} [ft]	h_{max} [ft]	WTC
BA11	direct	BAE	111	BA11	35000	29750	M
BDOG	equiv.	BAE	SA-3 BULLDOG	C182	20000		L
BE10	equiv.	BEECH	KING AIR 100	D228	28000		L
BE20	direct	BEECH	SUPER KING AIR 200	BE20	35000		L
BE30	equiv.	BEECH	SUPER KING AIR 300	BE20	35000		L
BE33	equiv.	BEECH	BONANZA 33	C182	20000		L
BE36	equiv.	BEECH	BONANZA 36	DA42	18000		L
BE40	equiv.	BEECH	BEECHJET 400	C25C	45000		M
BE55	equiv.	BEECH	BARON 55	PA34	25000	22500	L
BE58	direct	BEECH	BARON 58	BE58	25000		L
BE60	equiv.	BEECH	DUKE 60	C421	30000		L
BE65	equiv.	BEECH	QUEEN AIR 65	PA34	25000	22500	L
BE76	equiv.	BEECH	DUCHESSE 76	C182	20000		L
BE95	equiv.	BEECH	TRAVEL AIR 95	TRIN	25000		L
BE99	direct	BEECH	AIRLINER C99	BE99	15000		L
BE9L	direct	BEECH	KING AIR 90	BE9L	31000		L
BE9T	equiv.	BEECH	KING AIR 90	BE9L	31000		L
BMAN	equiv.	AAK	BUSHMAN	P28A	12000		L
BN2P	equiv.	BRITTEN-NORMAN	BN-2A MARITIME DEFENDER	PA34	25000	22500	L
BN2T	equiv.	BRITTEN-NORMAN	BN-2T Defender 4000	E120	32000		M
C10T	equiv.	ADVANCED AIRCRAFT	SPIRIT 750	D228	28000		L
C130	direct	LOCKHEED	HERCULES	C130	33000	21891	M
C135	equiv.	BOEING	STRATOLIFTER C-135C	B703	42000	35000	H
C141	equiv.	LOCKHEED	STARLIFTER C-141	A310	41000	35718	H
C160	direct	TRANSALL	C160	C160	30000	25500	M
C162	equiv.	CESSNA	SKYCATCHER	DA42	18000		L
C17	equiv.	BOEING	GLOBEMASTER 3	B764	45000	33210	H
C170	equiv.	CESSNA	170	C172	14000		L
C172	direct	CESSNA	SKYHAWK 172	C172	14000		L

A/C Code	Model Type	Aircraft manufacturer	Aircraft model	Synonym aircraft	h_{MO} [ft]	h_{max} [ft]	WTC
C177	equiv.	CESSNA	CARDINAL 177	C172	14000		L
C182	direct	CESSNA	SKYLANE 182	C182	20000		L
C206	equiv.	CESSNA	TURBO STATIONAIR 6	SR22	17500		L
C208	equiv.	CESSNA	CARAVAN	PA27	20000		L
C210	equiv.	CESSNA	CENTURION	SR22	17500		L
C212	equiv.	CASA	T-12 AVIOCAR	D228	28000		L
C25A	direct	CESSNA	525A Citation CJ2	C25A	45000		L
C25B	direct	CESSNA	525B Citation CJ3	C25B	45000		L
C25C	direct	CESSNA	525C Citation CJ4	C25C	45000		M
C27J	equiv.	ALENIA	C-27J Spartan	E120	32000		M
C295	equiv.	CASA	C-295	ATP	25000	21628	M
C303	equiv.	CESSNA	CRUSADER 303	PA31	26300		L
C30J	equiv.	LOCKHEED MARTIN	C130J HERCULES	C130	33000	21891	M
C310	equiv.	CESSNA	310	PA34	25000	22500	L
C337	equiv.	CESSNA	SUPER SKYMASTER	PA27	20000		L
C340	equiv.	CESSNA	C-340/340A	C421	30000		L
C402	equiv.	CESSNA	402	PA34	25000	22500	L
C404	equiv.	CESSNA	TITAN	BE58	25000		L
C414	equiv.	CESSNA	CHANCELLOR 414	PA31	26300		L
C421	direct	CESSNA	GOLDEN EAGLE 421	C421	30000		L
C425	equiv.	CESSNA	CORSAIR	TBM8	31000		L
C441	equiv.	CESSNA	Conquest	PAY3	33000		L
C5	equiv.	LOCKHEED	L-500 GALAXY	A346	41500	33613	H
C500	equiv.	CESSNA	CITATION 1	C550	43000		L
C501	equiv.	CESSNA	CITATION 1SP	C550	43000		L
C510	direct	CESSNA	CITATION MUSTANG	C510	41000		L
C525	direct	CESSNA	CITATION CJ1	C525	41000		L
C526	equiv.	CESSNA	CITATION JET 526	E50P	41000	40400	L
C550	direct	CESSNA	CITATION II-S2	C550	43000		L
C551	equiv.	CESSNA	CITATION 2SP	C550	43000		L
C560	direct	CESSNA	CITATION V	C560	45000	41516	M

A/C Code	Model Type	Aircraft manufacturer	Aircraft model	Synonym aircraft	h_{MO} [ft]	h_{max} [ft]	WTC
C56X	direct	CESSNA	CITATION Excel	C56X	45000	44523	M
C650	equiv.	CESSNA	CITATION VII	LJ35	45000	40287	M
C680	direct	CESSNA	Citation Sovereign	C680	47000		M
C72R	equiv.	CESSNA	CUTLASS RG	SR22	17500		L
C750	direct	CESSNA	CITATION 10	C750	51000	45180	M
C77R	equiv.	CESSNA	CARDINAL 177RG	P28A	12000		L
C82R	equiv.	CESSNA	SKYLANE R182 RG	C182	20000		L
CA41	equiv.	CORVUS	CA-41 RACER	P28A	12000		L
CE43	equiv.	CERVA	GUEPARD	SR22	17500		L
CL2T	equiv.	CANADAIR	CL-415	SH36	20000		M
CL30	equiv.	BOMBARDIER	Challenger 300	F2TH	47000	41486	M
CL60	direct	CANADAIR	CHALLENGER 600/601	CL60	41000	39223	M
CN35	equiv.	CASA	CN-235	AT43	25000	22699	M
COL4	equiv.	CESSNA	COLUMBIA 400	TRIN	25000		L
CORV	equiv.	WOLFSBERT	CORVUS	DA42	18000		L
CRJ1	direct	CANADAIR	REGIONAL JET CRJ-100	CRJ1	41000	34333	M
CRJ2	direct	CANADAIR	REGIONAL JET CRJ-200	CRJ2	41000	36855	M
CRJ7	equiv.	CANADAIR	REGIONAL JET CRJ-700	CRJ9	41000	36457	M
CRJ9	direct	CANADAIR	REGIONAL JET CRJ-900	CRJ9	41000	36457	M
CRJX	equiv.	BOMBARDIER	REGIONAL JET CRJ-1000	CRJ9	41000	36457	M
CVLT	equiv.	CANADAIR	CC-109 COSMOPOLITAN	DH8C	25000	24804	L
D228	direct	DORNIER	228	D228	28000		L
D328	direct	DORNIER	328	D328	32800	29051	M
DA40	equiv.	DIAMOND	DA-40-180 DIAMOND STAR	DA42	18000		L
DA42	direct	DIAMOND	TWIN STAR	DA42	18000		L
DC10	direct	MCDONNELL DOUGLAS	DC-10	DC10	42000	32000	H
DC3	equiv.	DOUGLAS	DC-3	C421	30000		L

A/C Code	Model Type	Aircraft manufacturer	Aircraft model	Synonym aircraft	h_{MO} [ft]	h_{max} [ft]	WTC
DC85	equiv.	MCDONNELL DOUGLAS	DC-8-50	A310	41000	35718	H
DC86	equiv.	MCDONNELL DOUGLAS	DC-8-60	DC87	42000	34000	H
DC87	direct	MCDONNELL DOUGLAS	DC-8-70	DC87	42000	34000	H
DC91	equiv.	MCDONNELL DOUGLAS	DC-9-10	B712	37000	35187	M
DC92	equiv.	MCDONNELL DOUGLAS	DC-9-20	DC94	35000	33500	M
DC93	equiv.	MCDONNELL DOUGLAS	DC-9-30	DC94	35000	33500	M
DC94	direct	MCDONNELL DOUGLAS	DC-9-40	DC94	35000	33500	M
DC95	equiv.	MCDONNELL DOUGLAS	DC-9-50	DC94	35000	33500	M
DH8A	direct	DE HAVILLAND	DASH 8-100	DH8A	25000	25000	M
DH8B	equiv.	DE HAVILLAND	DASH 8-200	DH8C	25000	24804	L
DH8C	direct	DE HAVILLAND	DASH 8-300	DH8C	25000	24804	L
DH8D	direct	DE HAVILLAND	DASH 8-400	DH8D	25000		M
DHC6	equiv.	DE HAVILLAND CANADA	DHC-6 Twin Otter	D228	28000		L
DR30	equiv.	ROBIN	PETIT PRINCE DR-315	PA44	15000	12527	L
DR40	equiv.	ROBIN	DR-400 DAUPHIN	PA44	15000	12527	L
DV2	equiv.	DOVA	DV-2 INFINITY	DA42	18000		L
E110	equiv.	EMBRAER	BANDEIRANTE	SW4	25000	25000	L
E120	direct	EMBRAER	EMB-120 BRASILIA	E120	32000		M
E121	equiv.	EMBRAER	Xingu	PAY2	29000		L
E135	direct	EMBRAER	EMB-135	E135	41000	38617	M
E145	direct	EMBRAER	EMB-145	E145	37000		M
E170	direct	EMBRAER	EMB-175	E170	41000		M
E190	direct	EMBRAER	EMB-190	E190	41000		M
E2	equiv.	GRUMMAN	E-2 HAWKEYE	E120	32000		M
E3CF	equiv.	BOEING	E-3 SENTRY	B762	43000	35861	H
E3TF	equiv.	BOEING	E-3A SENTRY	B762	43000	35861	H
E50P	direct	EMBRAER	Phenom 100	E50P	41000	40400	L
E55P	direct	EMBRAER	Phenom 300	E55P	45000	44923	M

A/C Code	Model Type	Aircraft manufacturer	Aircraft model	Synonym aircraft	h_{MO} [ft]	h_{max} [ft]	WTC
EA50	direct	ECLIPSE	ECLIPSE 500	EA50	41000		L
EPX1	equiv.	EPERVIER	X-1	DA42	18000		L
ETAR	equiv.	DASSAULT	ETENDARD 4	FGTN	50000		M
EUFI	equiv.	EUROFIGHTER	2000	FGTN	50000		M
EV97	equiv.	EVEKTOR	SPORTSTAR	P28A	12000		L
F1	equiv.	MITSUBISHI	F1	FGTN	50000		M
F100	direct	FOKKER	100	F100	35000	35000	M
F104	equiv.	LOCKHEED	STARFIGHTER	FGTN	50000		M
F117	equiv.	LOCKHEED	NIGHTHAWK	FGTN	50000		M
F14	equiv.	GRUMMAN	TOMCAT	FGTN	50000		M
F15	equiv.	MCDONNELL DOUGLAS	EAGLE	FGTN	50000		M
F16	equiv.	GENERAL DYNAMICS	FIGHTING FALCON	FGTN	50000		M
F18	equiv.	MCDONNELL DOUGLAS	HORNET	FGTN	50000		M
F1FV	equiv.	AVION	FAVORIT	TRIN	25000		L
F260	equiv.	SIAI-MARCHETTI	SF-260	BE58	25000		L
F27	direct	FOKKER	FRIENDSHIP	F27	25000	22777	M
F28	direct	FOKKER	FOLLOWSHIP	F28	35000	31000	M
F2TH	direct	DASSAULT	FALCON 2000	F2TH	47000	41486	M
F4	equiv.	MCDONNELL DOUGLAS	PHANTOM	FGTN	50000		M
F406	equiv.	CESSNA	F406 Vigilant	PAY3	33000		L
F5	equiv.	NORTHROP	F-5	FGTN	50000		M
F50	direct	FOKKER	50	F50	25000	22108	M
F5SA	equiv.	IRIAF	SAEGHE	FGTN	50000		M
F70	direct	FOKKER	70	F70	37000	36565	M
F900	direct	DASSAULT	FALCON 900	F900	51000	38187	M
FA04	equiv.	FLAMING AIR	PEREGRINE	P28A	12000		L
FA10	direct	DASSAULT	FALCON 10	FA10	45000	38400	M
FA20	direct	DASSAULT	FALCON 20	FA20	42000	38000	M
FA50	direct	DASSAULT	FALCON 50	FA50	49000	41177	M
FA7X	direct	DASSAULT	FALCON 7X	FA7X	51000	39848	M

A/C Code	Model Type	Aircraft manufacturer	Aircraft model	Synonym aircraft	h_{MO} [ft]	h_{max} [ft]	WTC
FGTH	direct	GENERIC	MIL FIGHTER HEAVY	FGTH	50000		M
FGTL	direct	GENERIC	MIL FIGHTER LIGHT	FGTL	41000		H
FGTN	direct	GENERIC	MIL FIGHTER NORMAL	FGTN	50000		M
G120	equiv.	GROB	G-120A	SR22	17500		L
G12T	equiv.	GROB	G-120TP	TBM7	31000		L
G150	equiv.	IAI	Gulfstream G150	F2TH	47000	41486	M
G222	equiv.	ALENIA	SPARTAN C-27A	ATP	25000	21628	M
G250	equiv.	GULFSTREAM	G250	F2TH	47000	41486	M
G280	equiv.	GULFSTREAM AEROSPACE	G280	F2TH	47000	41486	M
GA7	equiv.	GRUMMAN AMERICAN	COUGAR	SR22	17500		L
GALX	equiv.	IAI	1126 GALAXY	F2TH	47000	41486	M
GL5T	equiv.	BOMBARDIER	Global 5000	C750	51000	45180	M
GLAS	equiv.	STODDARD-HAMILTON	GLASAIR	BE58	25000		L
GLEX	equiv.	BOMBARDIER	BD-700 Global Express	C750	51000	45180	M
GLF2	equiv.	GULFSTREAM	GULFSTREAM II	FA7X	51000	39848	M
GLF3	equiv.	GULFSTREAM	GULFSTREAM III	FA7X	51000	39848	M
GLF4	equiv.	GULFSTREAM	GULFSTREAM IV	FA7X	51000	39848	M
GLF5	equiv.	GULFSTREAM	GULFSTREAM V C37	FA7X	51000	39848	M
GLF6	equiv.	GULFSTREAM	G650	FA7X	51000	39848	M
GLST	equiv.	GLASAIR	GlaStar	TRIN	25000		L
GRIZ	equiv.	AERO TEK	TURBO GRIZZLY	PA34	25000	22500	L
H25A	direct	HAWKER SIDDELEY	DOMINE HS 125	H25A	40000		M
H25B	direct	HAWKER BEECHCRAFT	HAWKER 800XP	H25B	43000	38507	M
H25C	equiv.	RAYTHEON	HAWKER 1000	H25B	43000	38507	M
HA4T	equiv.	RAYTHEON	HAWKER 4000	F2TH	47000	41486	M
HAR	equiv.	HAWKER SIDDELEY	HARRIER	FGTN	50000		M
HAWK	equiv.	HAWKER SIDDELEY	HAWK	FGTN	50000		M
HELI	equiv.	GENERIC	HELICOPTER	P28A	12000		L

A/C Code	Model Type	Aircraft manufacturer	Aircraft model	Synonym aircraft	h_{MO} [ft]	h_{max} [ft]	WTC
HR10	equiv.	ROBIN	TIARA	TRIN	25000		L
HRNT	equiv.	AAK	HORNET	P28A	12000		L
IL18	equiv.	ILYUSHIN	IL-18	C130	33000	21891	M
IL62	equiv.	ILYUSHIN	IL-62 /-62M / MK	A30B	39000	31966	H
IL76	direct	ILYUSHIN	IL-76TD	IL76	39700	28685	H
IL86	direct	ILYUSHIN	IL-86	IL86	37402	32337	H
IL96	direct	ILYUSHIN	IL-96-300	IL96	42979	31002	H
IMPX	equiv.	XTREMAIR	XTREME 3000	BE58	25000		L
J328	equiv.	FAIRCHILD DORNIER	328 Jet	E135	41000	38617	M
JAGR	equiv.	SEPECAT	JAGUAR	FGTN	50000		M
JS1	equiv.	JETSTREAM	JETSTREAM 1	BE20	35000		L
JS20	equiv.	JETSTREAM	JETSTREAM 200	D228	28000		L
JS31	equiv.	BAE	JETSTREAM 31	JS32	25000		L
JS32	direct	JETSTREAM	JETSTREAM Super 31	JS32	25000		L
JS41	direct	JETSTREAM	JETSTREAM 41	JS41	26000	24685	M
K35A	equiv.	BOEING	STRATOTANKER KC-135A	B703	42000	35000	H
K35E	equiv.	BOEING	STRATOTANKER KC-135D/E	B703	42000	35000	H
K35R	equiv.	BOEING	STRATOTANKER K35R	B703	42000	35000	H
KAT3	equiv.	KHRUNICHEV	AT-3	PA31	26300		L
KEST	equiv.	FARNBOROUGH	KESTREL	PAY2	29000		L
L101	direct	LOCKHEED	TRISTAR L-1011	L101	42000	33000	H
L159	equiv.	AERO (2)	L-159	FGTN	50000		M
L188	equiv.	LOCKHEED	ELECTRA L-188	C160	30000	25500	M
L29A	equiv.	LOCKHEED	JETSTART	CL60	41000	39223	M
L29B	equiv.	LOCKHEED	L1329 JETSTAR	CL60	41000	39223	M
L410	equiv.	LET	TURBOLET 410	D228	28000		L
LJ24	equiv.	LEARJET	24	C25A	45000		M
LJ25	equiv.	LEARJET	25	LJ45	51000	44099	M
LJ31	equiv.	LEARJET	31	LJ45	51000	44099	M
LJ35	direct	LEARJET	35	LJ35	45000	40287	M

A/C Code	Model Type	Aircraft manufacturer	Aircraft model	Synonym aircraft	h_{MO} [ft]	h_{max} [ft]	WTC
LJ40	equiv.	LEARJET	40	LJ45	51000	44099	M
LJ45	direct	LEARJET	45	LJ45	51000	44099	M
LJ55	equiv.	LEARJET	55	LJ45	51000	44099	M
LJ60	equiv.	LEARJET	60	LJ45	51000	44099	M
LNP4	equiv.	LANCAIR	PropJet 4	TBM8	31000		L
M2	equiv.	KUBICEK	M-2 SCOUT	P28A	12000		L
M20P	equiv.	MOONEY	MARK 201	TRIN	25000		L
M20T	equiv.	MOONEY	MARK 20T	TRIN	25000		L
MD11	direct	MCDONNELL DOUGLAS	MD-11	MD11	43000	31837	H
MD81	equiv.	MCDONNELL DOUGLAS	MD-81	MD82	37000	34448	M
MD82	direct	MCDONNELL DOUGLAS	MD-82	MD82	37000	34448	M
MD83	direct	MCDONNELL DOUGLAS	MD-83	MD83	37000	33513	M
MD87	equiv.	MCDONNELL DOUGLAS	MD-87	MD82	37000	34448	M
MD88	equiv.	MCDONNELL DOUGLAS	MD-88	MD82	37000	34448	M
MD90	equiv.	MCDONNELL DOUGLAS	MD-90	MD83	37000	33513	M
MG21	equiv.	MIKOYAN	MIG-21	FGTN	50000		M
MG23	equiv.	MIKOYAN	MIG-23	FGTN	50000		M
MG25	equiv.	MIKOYAN	MIG-25	FGTN	50000		M
MG29	equiv.	MIKOYAN	MIG-29	FGTN	50000		M
MIR2	equiv.	DASSAULT	MIRAGE 2000	FGTN	50000		M
MIR4	equiv.	DASSAULT	MIRAGE IV	FGTN	50000		M
MRF1	equiv.	DASSAULT	MIRAGE F1	FGTN	50000		M
MU2	direct	MITSUBISHI	MARQUISE / SOLITAIRE	MU2	28000		L
MU30	equiv.	MITSUBISHI	MU-300	C560	45000	41516	M
N262	equiv.	NORD	262	JS41	26000	24685	M
NIM	equiv.	HAWKER SIDDELEY	NIMROD	B738	41000	34982	M
NM5	equiv.	NAL	NM-5	TRIN	25000		L
NNJA	equiv.	BEST OFF	NYNJA	P28A	12000		L

A/C Code	Model Type	Aircraft manufacturer	Aircraft model	Synonym aircraft	h_{MO} [ft]	h_{max} [ft]	WTC
ONEX	equiv.	SONEW	ONEX	DA42	18000		L
P180	direct	PIAGGIO	P180 AVANTI	P180	41000		L
P210	equiv.	CESSNA	P210	BE58	25000		L
P28A	direct	PIPER	PA-28-140 CHEROKEE	P28A	12000		L
P28B	equiv.	PIPER	PA-28-236 DAKOTA	SR22	17500		L
P28R	equiv.	PIPER	PA-28R-201 ARROW	DA42	18000		L
P28T	direct	PIPER	PA-28RT-201T	P28T	20000	19000	L
P3	equiv.	LOCKHEED	ORION P-3	C130	33000	21891	M
P32R	equiv.	PIPER	PA-32R-301 SARATOGA SP	C182	20000		L
P32T	equiv.	PIPER	TURBO LANCE 2	TRIN	25000		L
P46T	equiv.	PIPER	Malibu Meridian	BE9L	31000		L
P68	equiv.	PARTENAVIA	P-68 Observer	PA27	20000		L
P68T	equiv.	PARTENAVIA	AP-68-TP-300 SPARTACUS	D228	28000		L
PA18	equiv.	PIPER	PA-18 SUPER CUB	PA34	25000	22500	L
PA23	equiv.	PIPER	APACHE PA23-150/160	PA27	20000		L
PA27	direct	PIPER	AZTEC PA23-235/250	PA27	20000		L
PA30	equiv.	PIPER	Twin Comanche	TRIN	25000		L
PA31	direct	PIPER	CHIEFTAIN	PA31	26300		L
PA32	equiv.	PIPER	PA-32 CHEROKEE SIX	PA44	15000	12527	L
PA34	direct	PIPER	PA-34-220T SENECA III	PA34	25000	22500	L
PA44	direct	PIPER	PA-44-180 SEMINOLE	PA44	15000	12527	L
PA46	direct	PIPER	PA-46-350P MALIBU MIRAGE	PA46	25000		L
PA47	equiv.	PIPER	PA-47	C510	41000		L
PAY1	equiv.	PIPER	PA-A-31T1-500 CHEYENNE I	PAY2	29000		L
PAY2	direct	PIPER	PA-31T-620 CHEYENNE II	PAY2	29000		L
PAY3	direct	PIPER	PA-42-720 CHEYENNE III	PAY3	33000		L

A/C Code	Model Type	Aircraft manufacturer	Aircraft model	Synonym aircraft	h_{MO} [ft]	h_{max} [ft]	WTC
PAY4	equiv.	PIPER	PA-42-1000 CHEYENNE 400	P180	41000		L
PC12	equiv.	PILATUS	PC-12 SPECTRE	BE9L	31000		L
PC6T	equiv.	PILATUS	PC-6C TURBO-PORTER	D228	28000		L
PC9	equiv.	PILATUS	PC-9	BE9L	31000		L
PRM1	equiv.	HAWKER BEECHCRAFT	390 Premier 1	C25A	45000		L
PRXT	equiv.	PRIVATE EXPLORER	T-EXPLORER	PA34	25000	22500	L
R722	equiv.	BOEING	727-200RE SUPER 27	B722	37000	33845	M
RFAL	equiv.	DASSAULT	RAFALE	FGTN	50000		M
RJ1H	direct	BAE	RJ-100 Avroliner	RJ1H	35000	30949	M
RJ70	equiv.	BAE	RJ-70 Avroliner	RJ85	35000	32166	M
RJ85	direct	BAE	RJ-85 Avroliner	RJ85	35000	32166	M
S601	equiv.	AEROSPATIAL	SB 601 CORVETTE	C525	41000		L
SB05	equiv.	SAAB	SAAB 105	C550	43000		L
SB20	direct	SAAB	SAAB 2000	SB20	31000		M
SB32	equiv.	SAAB	LANSEN	FGTN	50000		M
SB35	equiv.	SAAB	DRAKEN	FGTN	50000		M
SB37	equiv.	SAAB	VIGGEN	FGTN	50000		M
SB39	equiv.	SAAB	GRIPEN	FGTN	50000		M
SBR1	equiv.	ROCKWELL	SABRELINER	FA10	45000	38400	M
SD4	equiv.	TOMARK	VIPER	P28A	12000		L
SF34	direct	SAAB	SF 340	SF34	25000	25000	M
SH33	equiv.	SHORTS	SH3-330	SH36	20000		M
SH36	direct	SHORTS	SH3-360	SH36	20000		M
SHRK	equiv.	SHARK AERO	SHARK	TRIN	25000		L
SLK3	equiv.	SLICK	360	PA27	20000		L
SLK5	equiv.	SLICK	540	TRIN	25000		L
SNTA	equiv.	AIRSPORT	SONATA	P28A	12000		L
SORA	equiv.	ACS	100 SORA	P28A	12000		L
SR01	equiv.	EURODISPLAY	SR-01 MAGIC	DA42	18000		L
SR20	equiv.	CIRRUS	SR-20	SR22	17500		L

A/C Code	Model Type	Aircraft manufacturer	Aircraft model	Synonym aircraft	h_{MO} [ft]	h_{max} [ft]	WTC
SR22	direct	CIRRUS	SR-22	SR22	17500		L
SW18	equiv.	SKYWOOD	TEDDY SW18	P28A	12000		L
SW2	equiv.	SWEARINGEN	MERLIN II	SW4	25000	25000	L
SW3	equiv.	SWEARINGEN	MERLIN III	PAY3	33000		L
SW4	direct	SWEARINGEN	MERLIN IV	SW4	25000	25000	L
T10	equiv.	TNM-AVIA	T10 AVIA TOR	P28A	12000		L
T134	direct	TUPOLEV	TU134A-3	T134	39000	34764	M
T154	direct	TUPOLEV	TU154M	T154	41000	37285	M
T160	equiv.	TUPOLEV	TU160	B742	45000	33180	H
T204	direct	TUPOLEV	TU 204-300	T204	39700	35429	M
T50S	equiv.	SUKHOI	T-50 PAKFA	FGTN	50000		M
TB30	equiv.	SOCATA	EPSILON TB30	PA27	20000		L
TBM7	direct	SOCATA	TBM-700	TBM7	31000		L
TBM8	direct	SOCATA	TBM-850	TBM8	31000		L
TEX2	equiv.	HAWKER BEECHCRAFT	T-6 TEXAN 2	TBM7	31000		L
TOBA	equiv.	SOCATA	TOBAGO TB-10	C172	14000		L
TOR	equiv.	PANAVIA	TORNADO	FGTN	50000		M
TRIN	direct	SOCATA	TRINIDAD TB-20	TRIN	25000		L
TRIS	equiv.	BRITTEN-NORMAN	Trislander	DA42	18000		L
TUCA	equiv.	EMBRAER	TUCANO	BE9L	31000		L
VC10	equiv.	VICKERS	VC10	B762	43000	35861	H
WSP	equiv.	AAK	WASP	P28A	12000		L
WW24	equiv.	IAI	1124 WESTWIND	FA10	45000	38400	M
YK40	direct	YAKOVLEV	YAK-40	YK40	26575		M
YK42	direct	YAKOVLEV	YAK-42	YK42	31496		M

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APPENDIX B

SOLUTIONS FOR BUFFETING LIMIT ALGORITHM

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A general solution for finding the roots of a cubic expression can be found in [RD15]. If we take expression 3.6-6, we can rewrite it to:

$$\mathrm{M}^3 - \frac{C_{Lbo(M=0)}}{k} \cdot \mathrm{M}^2 + \frac{\frac{W}{S}}{0.583 \cdot P \cdot k} = 0$$

Let:

$$a_1 = -\frac{C_{Lbo(M=0)}}{k}$$

$$a_2 = 0$$

$$a_3 = \frac{\frac{W}{S}}{0.583 \cdot P \cdot k}$$

Now let:

$$Q = \frac{\left(3 \cdot a_2 - a_1^{\,2}\right)}{9}$$

and:

$$R = \frac{\left(9 \cdot a_1 \cdot a_2 - 27 \cdot a_3 - 2 \cdot a_1^{\,3}\right)}{54}$$

The discriminant D is equal to: $Q^3 + R^2$. In our case D is usually strictly negative, which means that all roots are unequal and real. A simplified computation method with the help of trigonometry is given below:

$$\mathrm{X}_1 = 2 \cdot \sqrt{-Q} \cdot \cos\left(\frac{\theta}{3}\right) - \frac{a_1}{3}$$

$$\mathrm{X}_2 = 2 \cdot \sqrt{-Q} \cdot \cos\left(\frac{\theta}{3} + 120^0\right) - \frac{a_1}{3}$$

$$\mathrm{X}_3 = 2 \cdot \sqrt{-Q} \cdot \cos\left(\frac{\theta}{3} + 240^0\right) - \frac{a_1}{3}$$

With: $\cos\theta = \frac{R}{\sqrt{-Q^3}}$

The solutions X_1, X_2 and X_3 now give the possible values of M. One solution (in our case usually X_1) is always negative. The others are positive with the lower one (usually X_2) being the low speed buffeting limit we are looking for.

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